

SoC 를 위한 Peripheral 설계

[Create and Package New IP (I , II)]

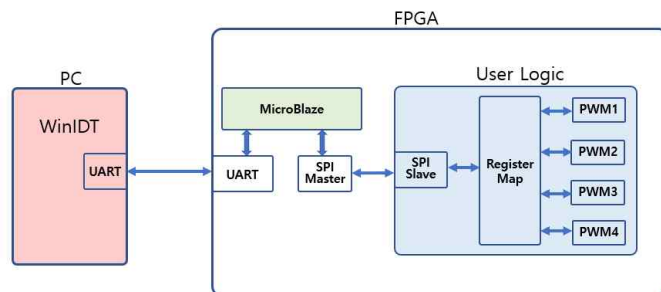
Reference :

2025-06-18

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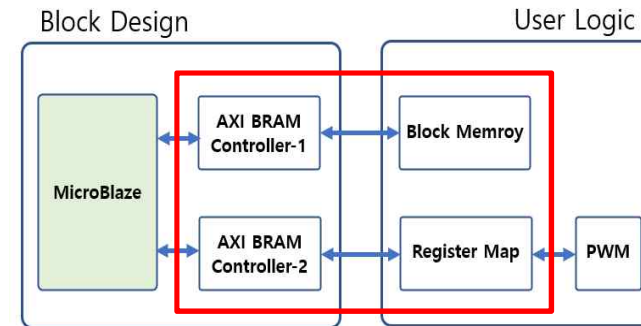


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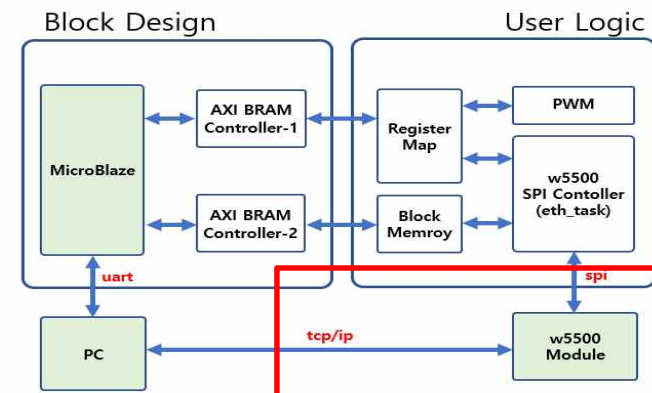
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➤ SoC Peripheral RTC Design Project

[Create and Package New IP (I)]

SPI_Task and Top_FndController

2025-06-18

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤ U0 : spi_task(spi_task.v)

spi_task(spi_task.v)(1/4)

```

1. `timescale 1ns / 1ps

2. module spi_task(reset , clock , btn , led
3. );
4.   input                reset ;
5.   input                clock ;
6.   input [3:0]          btn ;
7.   output [7:0]         led ;

8.   wire [3:0]          btn2 ;
9.   btn_in btn_in_u1 (
10.     .reset (reset ),
11.     .clock (clock ),
12.     .btn_in (btn[0] ),
13.     .btn_out (btn2[0] )
14.   );
15.   btn_in btn_in_u2 (
16.     .reset (reset ),
17.     .clock (clock ),
18.     .btn_in (btn[1] ),
19.     .btn_out (btn2[1] )
20.   );
21.   btn_in btn_in_u3 (
22.     .reset (reset ),
23.     .clock (clock ),
24.     .btn_in (btn[2] ),
25.     .btn_out (btn2[2] )
26.   );
27.   btn_in btn_in_u4 (
28.     .reset (reset ),
29.     .clock (clock ),
30.     .btn_in (btn[3] ),
31.     .btn_out (btn2[3] )
32.   );

33.   reg [1:0] btn_id;
34.   always @(posedge clock or negedge reset)
35.   begin
36.     if(~reset) btn_id <= 2'b0;
37.     else btn_id <= btn2[0] ? 2'b00 :
38.       btn2[1] ? 2'b01 :
39.       btn2[2] ? 2'b10 :
40.       btn2[3] ? 2'b11 : btn_id;
41.   end

42.   reg [7:0] wdata0;
43.   always @(posedge clock or negedge reset)
44.   begin
45.     if(~reset) wdata0 <= 8'h00;
46.     else wdata0 <= btn2[0] ? wdata0+1'b1 : wdata0 ;
47.   end

```

spi_task(spi_task.v)(2/4)

```

48.   reg [7:0] wdata1;
49.   always @(posedge clock or negedge reset)
50.   begin
51.     if(~reset) wdata1 <= 8'h00;
52.     else wdata1 <= btn2[1] ? wdata1+1'b1 : wdata1 ;
53.   end

54.   reg [7:0] wdata2;
55.   always @(posedge clock or negedge reset)
56.   begin
57.     if(~reset) wdata2 <= 8'h00;
58.     else wdata2 <= btn2[2] ? wdata2+1'b1 : wdata2 ;
59.   end

60.   reg [7:0] wdata3;
61.   always @(posedge clock or negedge reset)
62.   begin
63.     if(~reset) wdata3 <= 8'h00;
64.     else wdata3 <= btn2[3] ? wdata3+1'b1 : wdata3 ;
65.   end

66.   // -----
67.   parameter SLAVE_IDW = 8'h64;
68.   parameter SLAVE_IDR = 8'h65;
69.   parameter REG_ADDR1 = 8'h10;
70.   parameter REG_ADDR2 = 8'h11;
71.   parameter REG_ADDR3 = 8'h12;
72.   parameter REG_ADDR4 = 8'h13;
73.   parameter SPI_FREQ = 10'd16;

74.   // -----
75.   // 1) define state
76.   reg [2:0] m_state ;
77.   parameter M_IDLE = 3'd0 ;
78.   parameter M_WREADY = 3'd1 ;
79.   parameter M_WRITE = 3'd2 ;
80.   parameter M_RREADY = 3'd3 ;
81.   parameter M_READ = 3'd4 ;

82.   // -----
83.   // 2) state flag
84.   wire s_idle = (m_state == M_IDLE ) ? 1'b1 : 1'b0 ;
85.   wire s_wready = (m_state == M_WREADY ) ? 1'b1 : 1'b0 ;
86.   wire s_write = (m_state == M_WRITE ) ? 1'b1 : 1'b0 ;
87.   wire s_rready = (m_state == M_RREADY ) ? 1'b1 : 1'b0 ;
88.   wire s_read = (m_state == M_READ ) ? 1'b1 : 1'b0 ;

89.   // -----
90.   // 3) code implementation
91.   reg [1:0] wready_cnt;
92.   always @(posedge clock or negedge reset)
93.   begin
94.     if(~reset) wready_cnt <= 2'b0;
95.     else wready_cnt <= ~s_wready ? 2'b0 : wready_cnt+1'b1;
96.   end

```

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤ U0 : spi_task(spi_task.v)

spi_task(spi_task.v)(3/4)

```
97. wire [7:0] reg_addr = (btn_id==2'd0) ? REG_ADDR1 :
98. (btn_id==2'd1) ? REG_ADDR2 :
99. (btn_id==2'd2) ? REG_ADDR3 : REG_ADDR4 ;
100. reg [7:0] addr;
101. always @(posedge clock or negedge reset)
102. begin
103. if(~reset) addr <= 8'b0;
104. else addr <= (s_wready & (wready_cnt==2'b11)) ? reg_addr : addr ;
105. end

106. wire [7:0] reg_wdata = (btn_id==2'd0) ? wdata0 :
107. (btn_id==2'd1) ? wdata1 :
108. (btn_id==2'd2) ? wdata2 : wdata3 ;
109. reg [7:0] wdata;
110. always @(posedge clock or negedge reset)
111. begin
112. if(~reset) wdata <= 8'b0;
113. else wdata <= (s_wready & (wready_cnt==2'b11)) ? reg_wdata :
wdata ;
114. end
115.
116. reg start_w;
117. always @(posedge clock or negedge reset)
118. begin
119. if(~reset) start_w <= 1'b0;
120. else start_w <= (s_wready & (wready_cnt==2'b11)) ? 1'b1 : 1'b0 ;
121. end

122. reg [3:0] write_cnt;
123. always @(posedge clock or negedge reset)
124. begin
125. if(~reset) write_cnt <= 4'b0;
126. else write_cnt <= ~s_write ? 4'b0 :
127. (write_cnt==4'd15) ? 4'd15 : write_cnt+1'b1;
128. end

129. reg [1:0] rready_cnt;
130. always @(posedge clock or negedge reset)
131. begin
132. if(~reset) rready_cnt <= 2'b0;
133. else rready_cnt <= ~s_rready ? 2'b0 : rready_cnt+1'b1;
134. end

135. reg start_r;
136. always @(posedge clock or negedge reset)
137. begin
138. if(~reset) start_r <= 1'b0;
139. else start_r <= (s_rready & (rready_cnt==2'b11)) ? 1'b1 : 1'b0 ;
140. end
```

spi_task(spi_task.v)(4/4)

```
141. reg [3:0] read_cnt;
142. always @(posedge clock or negedge reset)
143. begin
144. if(~reset) read_cnt <= 4'b0;
145. else read_cnt <= ~s_read ? 4'b0 :
146. (read_cnt==4'd15) ? 4'd15 : read_cnt+1'b1;
147. end

148. wire [7:0] rdata ;
149. wire ss ;
150. wire sck ;
151. wire mosi ;
152. wire miso ;
153. wire done_mst;
154. spi_master spi_master_u1(.reset (reset ),.clock (clock ),.freq (SPI_FREQ ),
155. .start_w (start_w ),.start_r (start_r ),.addr (addr ),
156. .wdata (wdata ),.rdata (rdata ),.done (done_mst ),
157. .ss (ss ),.sck (sck ),.mosi (mosi ),
158. .miso (miso ));
159.

160. spi_slave spi_slave_u1(.reset (reset ),.clock (clock ),.ss (ss ),
161. .sck (sck ),.mosi (mosi ),.miso (miso ));
162.

163. reg [7:0] led;
164. always @(posedge clock or negedge reset)
165. begin
166. if(~reset) led <= 4'b0;
167. else led <= (s_read & (read_cnt==4'd15) & done_mst) ? rdata : led;
168. end

169. // -----
170. // 4) state transition
171. always @(posedge clock or negedge reset)
172. begin
173. if(~reset) m_state <= 3'b0;
174. else m_state <= (s_idle & (btn2 != 4'b0)) ? M_WREADY :
175. (s_wready & (wready_cnt==2'b11)) ? M_WRITE :
176. (s_write & (write_cnt==4'd15) & done_mst) ? M_RREADY :
177. (s_rready & (rready_cnt==2'b11)) ? M_READ :
178. (s_read & (read_cnt==4'd15) & done_mst) ? M_IDLE :
179. m_state ;
180. end

181. endmodule
```

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤ top_fndcontrol2

top_fndcontrol2(1/2)

```
1. `timescale 1ns / 1ps
2. module top_fndcontrol2(
3.     input clk,
4.     input reset,
5.     input mode,
6.     input [3:0] btn,
7.     output [3:0] fndsel,
8.     output [6:0] fnd,
9.     output [7:0] led_out
10. );
11. wire [7:0] led;
12. wire w_clkout;
13. wire [1:0] w_counter;
14. wire [3:0] w_datafnd;
15. wire [3:0] w_outa, w_outb, w_outc, w_outd;
16. wire [3:0] leda;
17. wire [3:0] ledb;
18. assign w_outa = {3'b000,btn[0]};
19. assign w_outb = {2'b00,btn[1],1'b0};
20. assign w_outc = {1'b0,btn[2],2'b00};
21. assign w_outd = {btn[3],3'b000};
22. assign leda = led[3:0];
23. assign ledb = led[7:4];
24. assign led_out = {ledb, leda};
25. spi_task U0(
26.     .reset(reset) ,
27.     .clock(clk) ,
28.     .btn(btn) ,
29.     .led(led)
30. );
```

top_fndcontrol2(2/2)

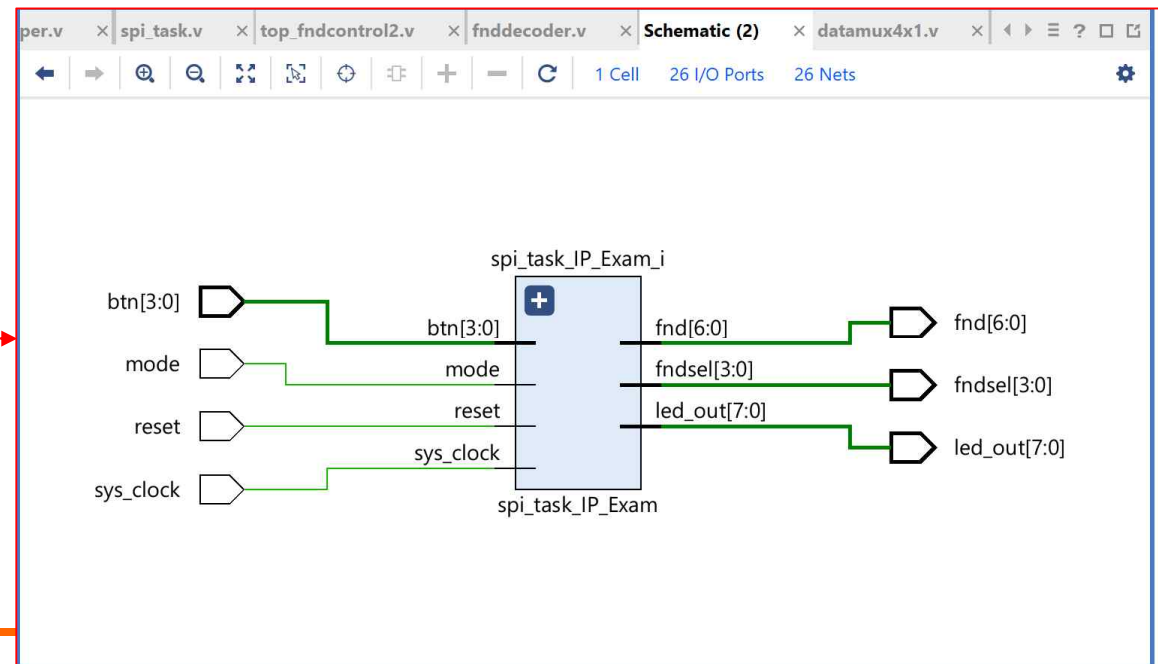
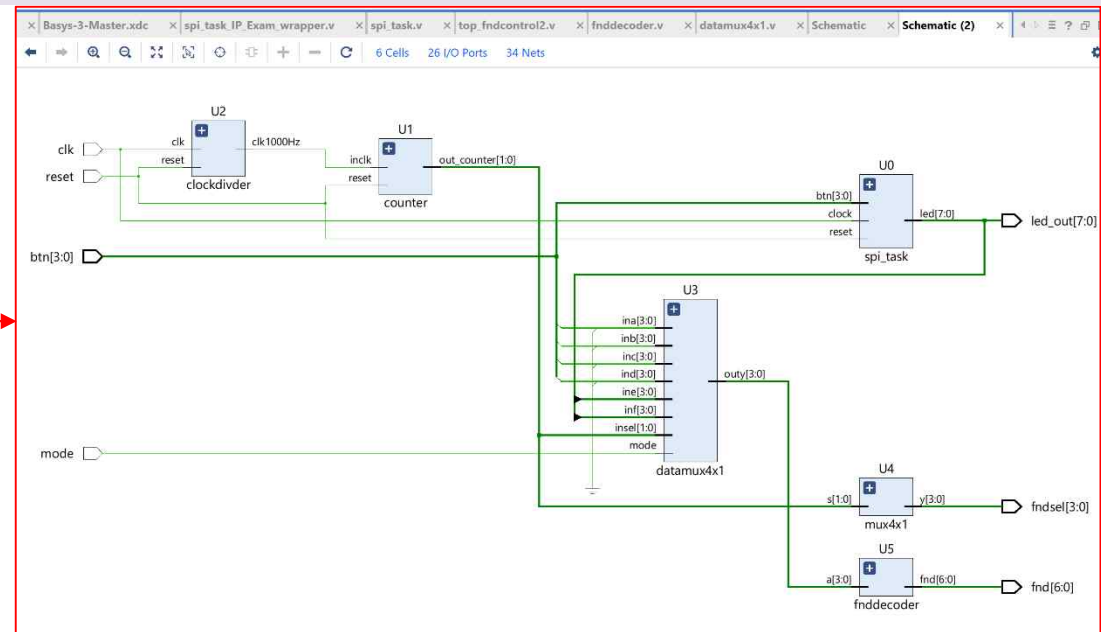
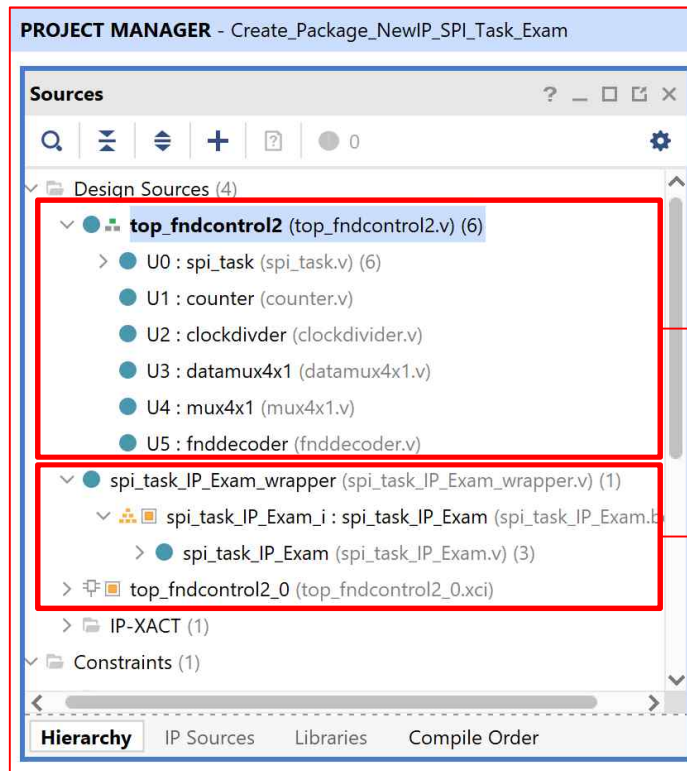
```
31. counter U1 (
32.     .inclk(w_clkout),
33.     .reset(reset),
34.     .out_counter(w_counter)
35. );
36. clockdivider U2 (
37.     .clk(clk),
38.     .reset(reset),
39.     .clk1000Hz(w_clkout)
40. );
41. datamux4x1 U3 (
42.     .mode(mode),
43.     .ina (w_outa),
44.     .inb (w_outb),
45.     .inc (w_outc),
46.     .ind (w_outd),
47.     .ine (leda),
48.     .inf (ledb),
49.     .insel(w_counter),
50.     .outy (w_datafnd)
51. );
52. mux4x1 U4 (
53.     .s(w_counter),
54.     .y(fndsel)
55. );
56. fnddecoder U5 (
57.     .reset(reset),
58.     .a(w_datafnd),
59.     .fnd(fnd)
60. );
61. endmodule
```

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

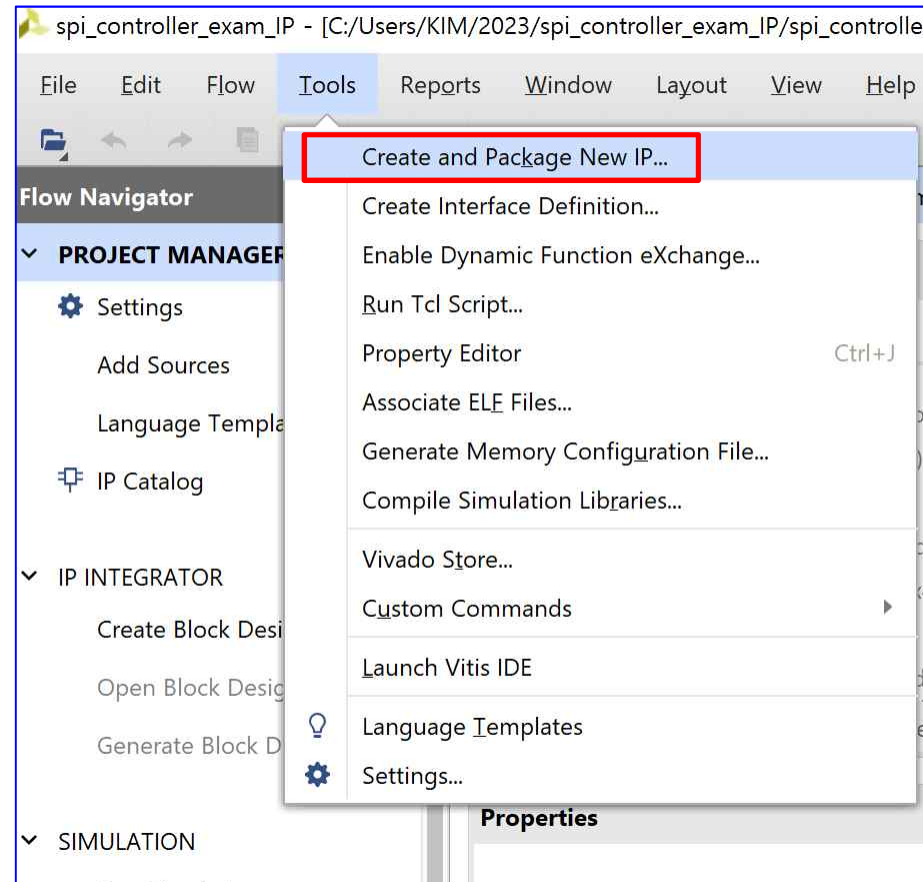
➤ spi_task and top_fndcontrol2

- Design Source → Create or Add



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



The screenshot shows the Vivado 2023.1 IDE interface. On the left, the 'Flow Navigator' pane is visible, showing the project structure. The 'PROJECT MANAGER' pane displays the 'Sources' tab, listing design sources for 'top_fndcontrol2'. The 'Create and Package New IP' wizard is open in the center, with the title bar 'Create and Package New IP'. The wizard content includes the AMD Vivado ML Edition logo and three tasks listed in a box:

- Create and Package New IP**
This wizard can be used to accomplish following tasks:
- Package a new IP for the Vivado IP Catalog**
This wizard will guide you through the process of creating a new Vivado IP using source files and information from your current project, block design or specified directory.
- Create a new AXI4 Peripheral**
This wizard will guide you through the process of creating a new AXI4 peripheral which includes HDL, driver, software test application, IP Integrator VIP simulation and debug demonstration design.

At the bottom of the wizard, there are buttons for '< Back', 'Next >', 'Finish', and 'Cancel'. The 'Next >' button is highlighted with a red box.

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Flow Navigator

PROJECT MANAGER - Create_Package_NewIP_Task_Exam

Sources

- Design Sources (1)
 - top_fndcontrol2 (top_fndcontrol2.v) (6)
 - U0 : spi_task (spi_task.v)
 - U1 : counter (counter.v)
 - U2 : clockdivider (clockdivider.v)
 - U3 : datamux4x1 (datamux4x1.v)
 - U4 : mux4x1 (mux4x1.v)
 - U5 : fnddecoder (fnddecoder.v)
- Constraints (1)
 - constrs_1 (1)
 - Basys-3-Master.xdc
- Simulation Sources (5)

Hierarchy Libraries Compile Order

Source File Properties

top_fndcontrol2.v

General Properties

Tcl Console Messages Log Reports Design

Create and Package New IP

Create Peripheral, Package IP or Package a Block Design

Please select one of the following tasks.

Packaging Options

- ☒ Package your current project
Use the project as the source for creating a new IP Definition.
- ☐ Package a block design from the current project
Choose a block design as the source for creating a new IP Definition.
- ☐ Package a specified directory
Choose a directory as the source for creating a new IP Definition.

Create AXI4 Peripheral

- ☐ Create a new AXI4 peripheral
Create an AXI4 IP, driver, software test application, IP Integrator AXI4 VIP simulation and debug demonstration design.

< Back Next > Finish Cancel

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

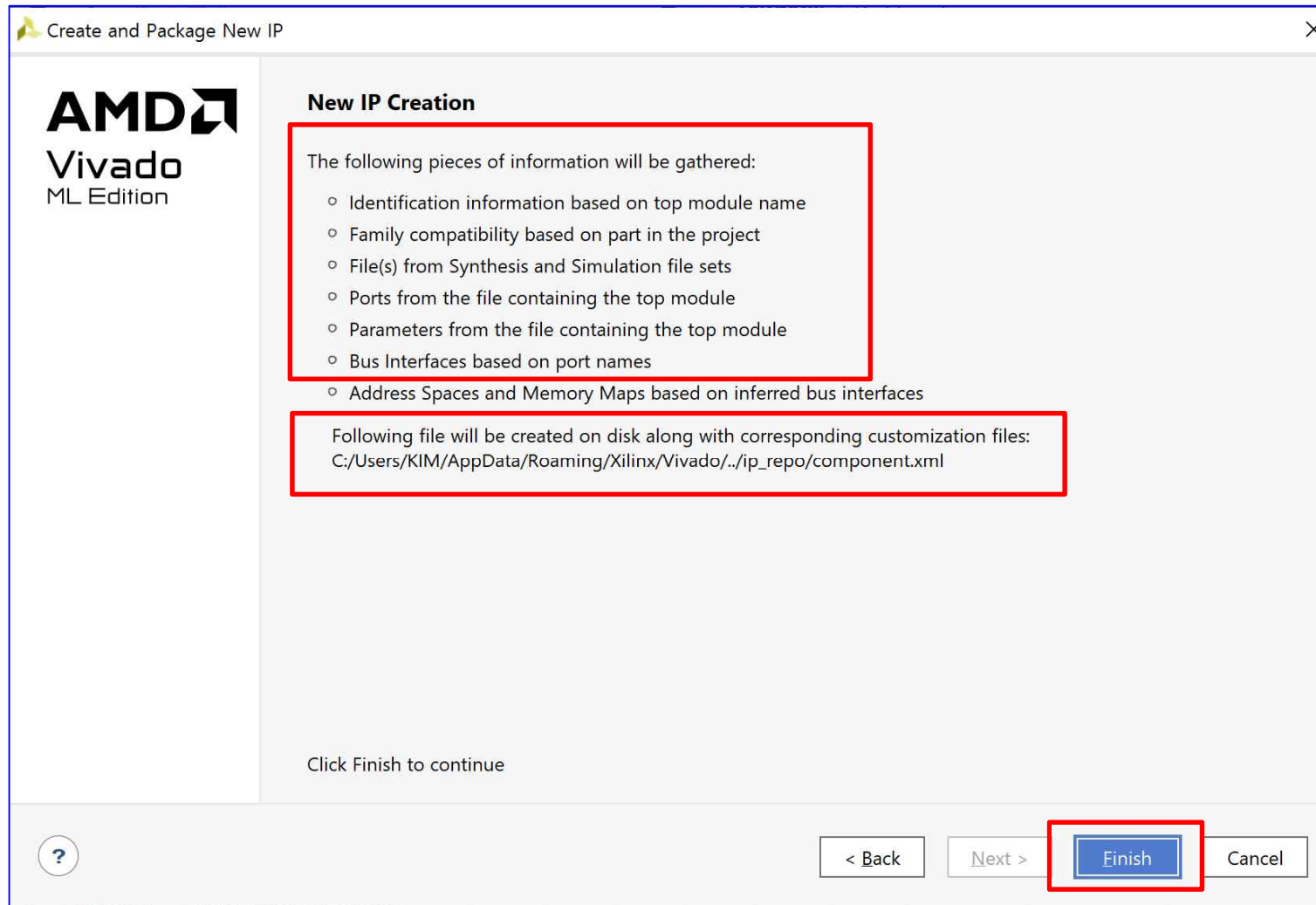
The image shows two windows from the Xilinx Vivado IDE. The left window is titled 'Create and Package New IP' and contains the 'Package Your Current Project' section. It instructs the user to 'Select the directory where the IP Definition will be created and the associated options for packaging the current project.' A text field labeled 'IP location:' contains the path 'C:/Users/KIM/AppData/Roaming/Xilinx/Vivado/./ip_repo'. A red box highlights this field, and a red arrow points from it to the right window. The right window is a file explorer titled 'IP Definition Source Location'. It shows a tree view of the file system. The 'Recent' list at the top includes 'C:/Users/KIM/AppData/Roaming/Xilinx/Vivado/./ip_repo'. The 'Directory:' field also shows this path. The tree view shows the following structure:

- Users
 - Default
 - KIM
 - .Xilinx
 - .eclipse
 - .vscode
 - 3D Objects
 - 2023
 - 2023_1
 - AppData
 - Local
 - LocalLow
 - Roaming
 - ATI
 - Adobe
 - AlphaX
 - Code
 - Digilent
 - ESTsoft
 - HNC
 - Microsoft
 - Mozilla
 - Naver
 - NuGet
 - Printmade3
 - SafedotX
 - Teams
 - Visual Studio Setup
 - Xilinx
 - Common
 - Vivado
 - Webtalk
 - ip_repo
 - updatemem

- Contacts
- Desktop
- Documents
- Downloads
- Favorites
- Intel

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



The screenshot shows the Vivado 2023.1 interface. The 'PROJECT MANAGER' tab is active, displaying the 'top_fndcontrol2' project. The 'Sources' pane shows the project hierarchy with components U0 through U5. The 'IP Packager' dialog is open, showing the 'Packaging Sources...' step. The 'Design Runs' table at the bottom shows the completion of synthesis and implementation.

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM
synth_1	constrs_1	synth_design Complete!									Partially routed, unrouted and conflict nets			343	354	0
impl_1	constrs_1	write_bitstream Complete!	5.205	0.000	0.060	0.000		0.000	0.097	0	2 CW, 20 Warn			339	362	0

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



PROJECT MANAGER - spi_controller_exam_IP

Project Summary × Package IP - top_fndcontrol2 ×

Packaging Steps

- ✓ Identification
- ✓ Compatibility
- ✓ File Groups
- Customization Parameters
- Ports and Interfaces
- Addressing and Memory
- ✓ Customization GUI
- Review and Package

Identification

Vendor: xilinx.com

Library: user

Name: top_fndcontrol2

Version: 1.0

Display name: top_fndcontrol2_v1_0

Description: top_fndcontrol2_v1_0

Vendor display name:

Company url:

Root directory: c:/Users/KIM/AppData/Roaming/Xilinx/ip_repo

Xml file name: c:/Users/KIM/AppData/Roaming/Xilinx/ip_repo/component.xml

Categories

+ - ↑ ↓

/UserIP

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



The screenshot shows the Xilinx Project Manager interface for a project named 'spi_controller_exam_IP'. The 'Package IP - top_fndcontrol2' tab is active. The left sidebar shows the 'Packaging Steps' with 'Review and Package' selected. The main area displays the 'Review and Package' summary, including the display name 'top_fndcontrol2_v1_0', description 'top_fndcontrol2_v1_0', and root directory 'c:/Users/KIM/AppData/Roaming/Xilinx/ip_repo', which is highlighted with a red box. Below the summary, the 'After Packaging' section states that an archive will not be generated and provides a link to 'Edit packaging settings'. At the bottom right, the 'Package IP' button is highlighted with a red box.

PROJECT MANAGER - spi_controller_exam_IP

Project Summary x Package IP - top_fndcontrol2 x

Packaging Steps

- ✓ Identification
- ✓ Compatibility
- ✓ File Groups
- Customization Parameters
- Ports and Interfaces
- Addressing and Memory
- ✓ Customization GUI
- Review and Package**

Review and Package

2 warnings 2 info messages

Summary

Display name: top_fndcontrol2_v1_0
Description: top_fndcontrol2_v1_0
Root directory: c:/Users/KIM/AppData/Roaming/Xilinx/ip_repo

After Packaging

An archive will not be generated. Use the settings link below to change your preference
IP will be made available in the catalog using the repository -
c:/Users/KIM/AppData/Roaming/Xilinx/ip_repo
[Edit packaging settings](#)

Package IP

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

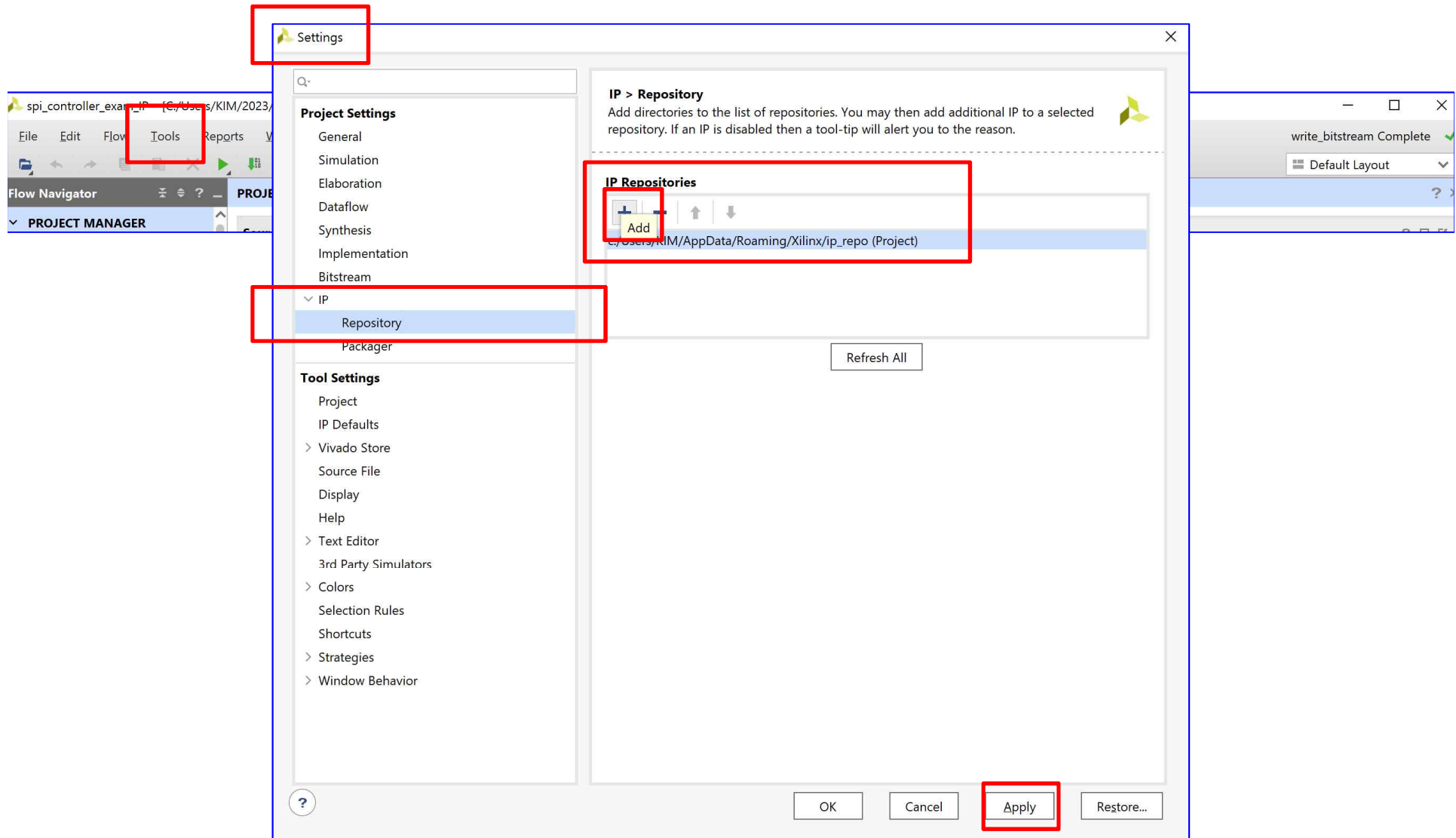


The screenshot displays the Vivado 2023.1 IDE interface. The top menu bar includes File, Edit, Flow, Tools, Reports, Window, Layout, View, and Help. The toolbar shows various icons for file operations and project management. The left sidebar contains the Flow Navigator and Project Manager. The Project Manager shows the project hierarchy for 'spi_controller_exam_IP'. The main workspace is divided into three panes: Sources, Properties, and Project Summary. The Sources pane shows the design sources for 'top_fndcontrol2', including 'U0 : spi_task', 'U1 : counter', 'U2 : clockdivider', 'U3 : datamux4x1', and 'U4 : mux4x1'. The Properties pane is empty. The Project Summary pane shows the 'Package IP - top_fndcontrol2' process, which is currently in the 'Review and Package' step. The 'Review and Package' step shows a summary of the IP, including the display name 'top_fndcontrol2_v1_0', description 'top_fndcontrol2_v1_0', and root directory 'c:/Users/KIM/AppData/Roaming/Xilinx/ip_repo'. The 'After Packaging' section indicates that an archive will not be generated and provides a link to 'Edit packaging settings'. A 'Re-Package IP' button is visible at the bottom right of the Project Summary pane.

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

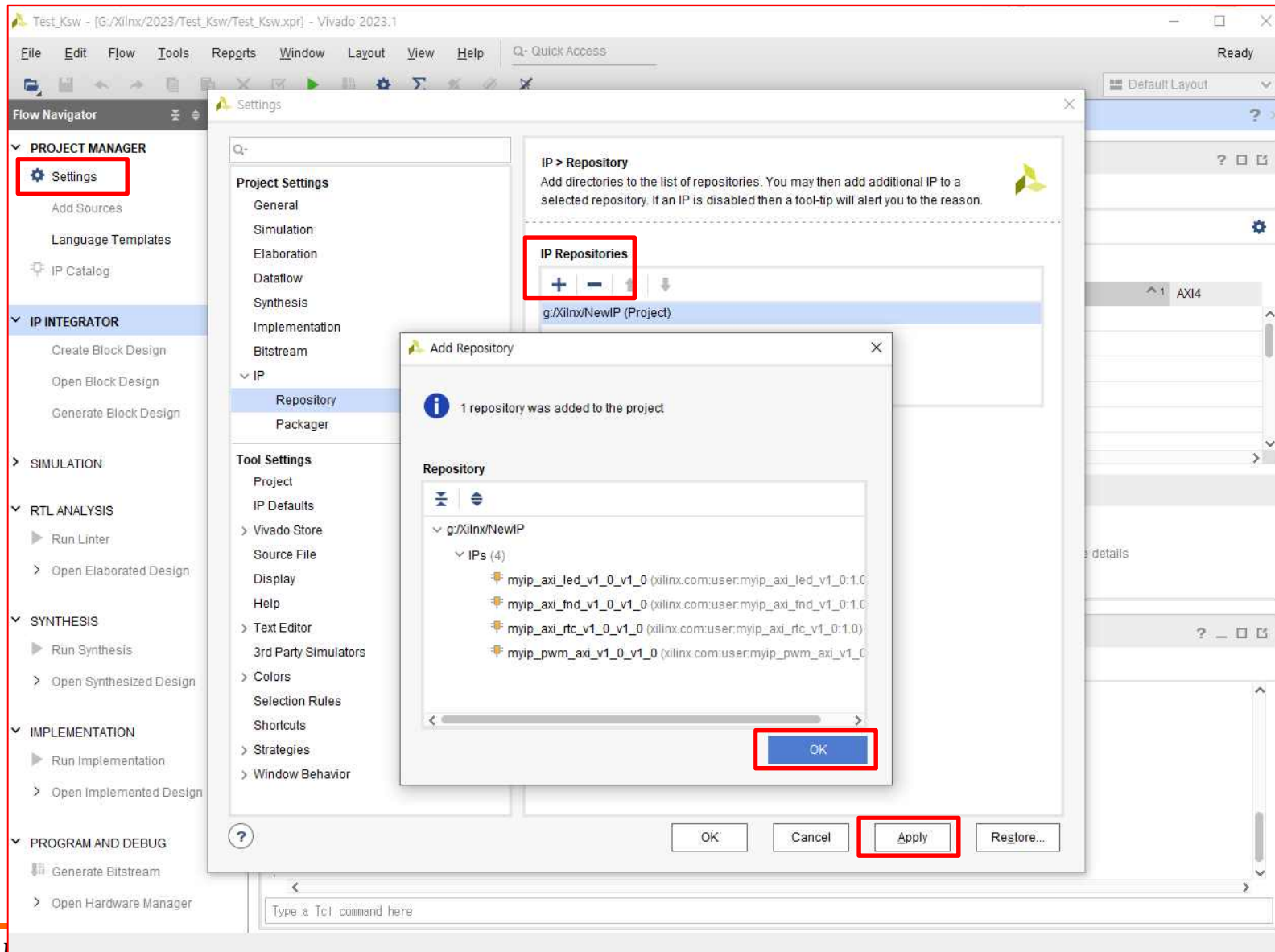
➤ **IP ➔ ADD** : Tools ➔ Setting ➔ IP ➔ Repository ➔ ADD



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

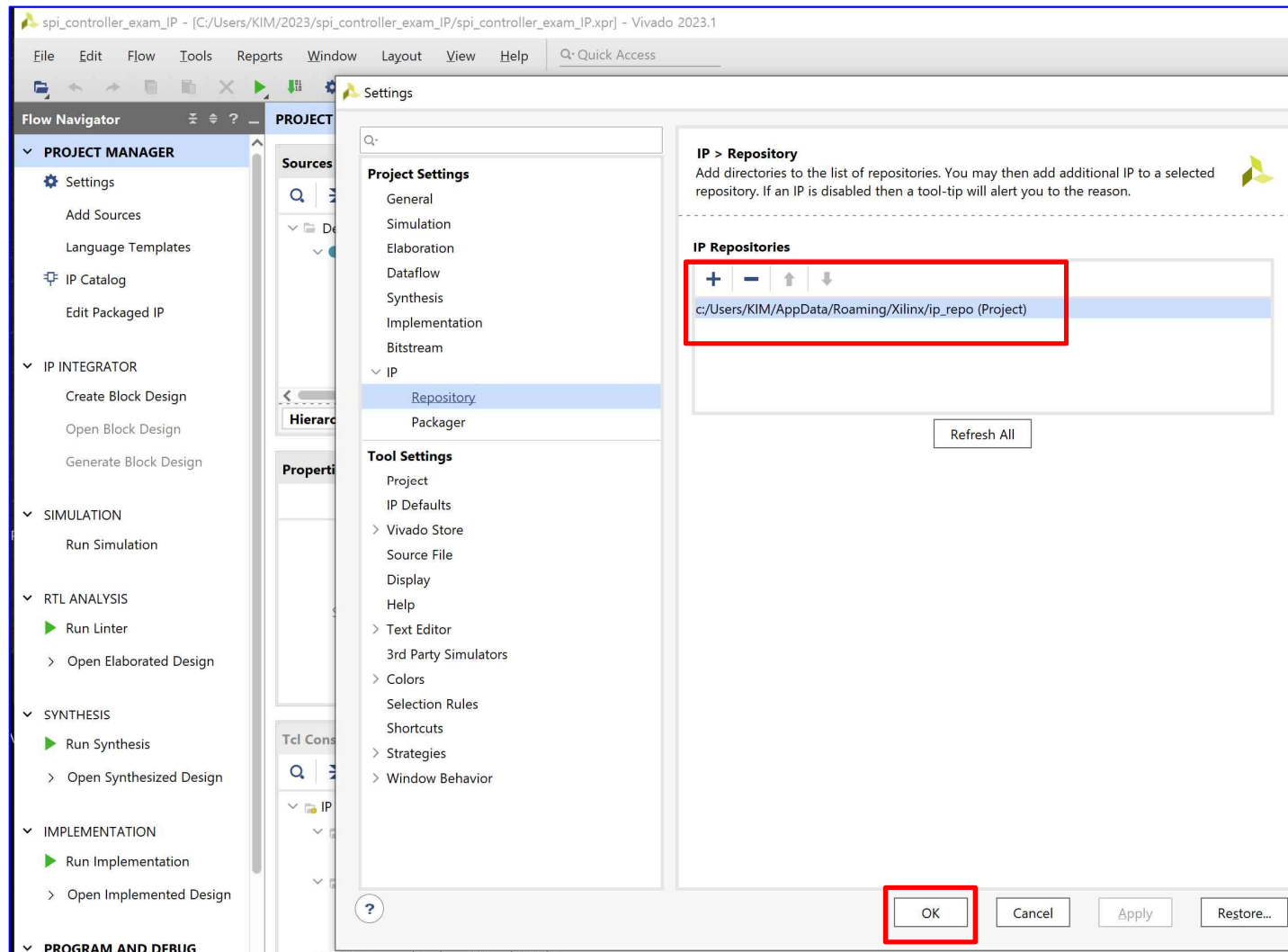
➤ **IP ➔ ADD** : Tools ➔ Setting ➔ IP ➔ Repository ➔ ADD



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
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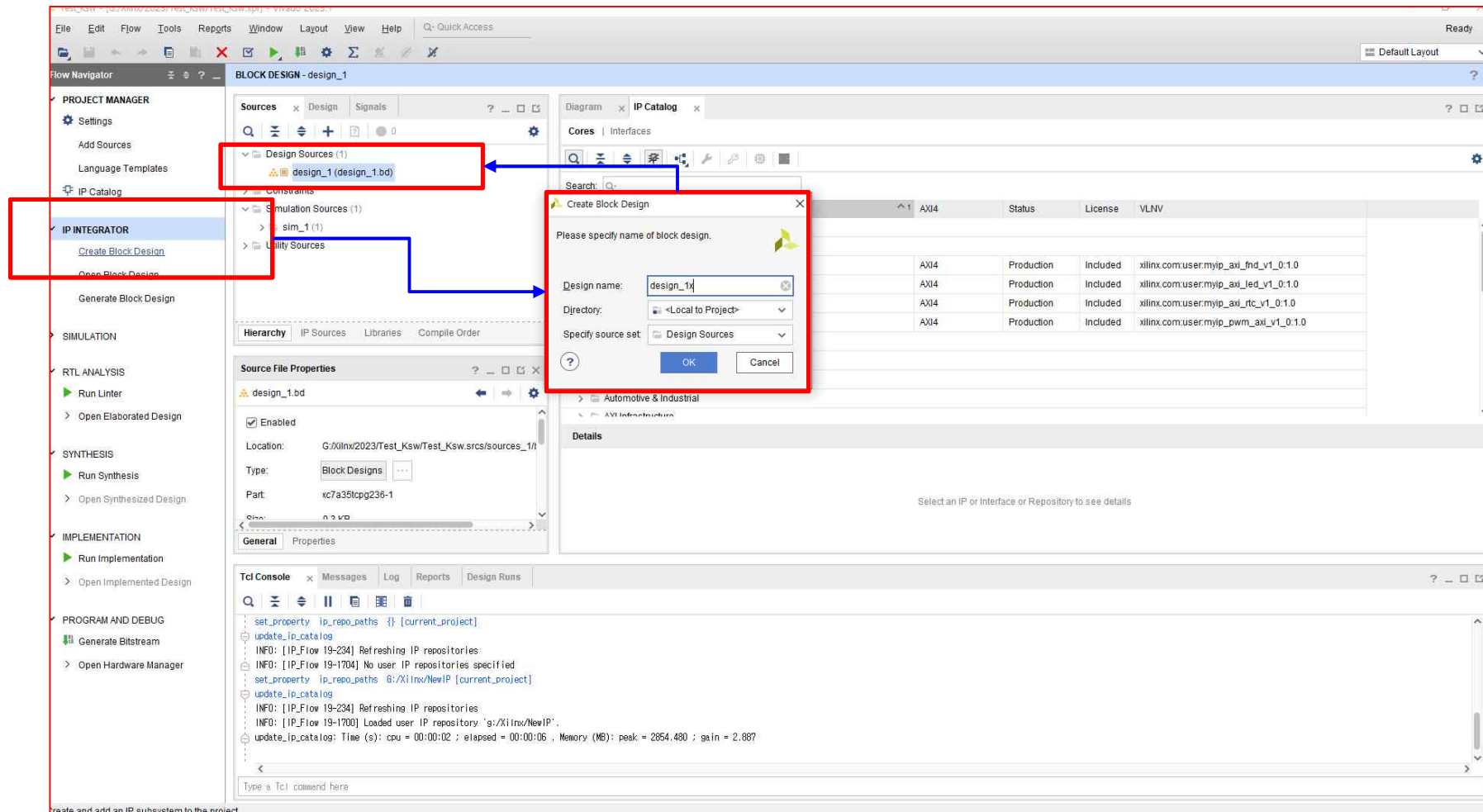
➤ IP ➔ ADD : Tools ➔ Setting ➔ IP ➔ Repository ➔ ADD



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
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➤ IP INTEGRATOR ➔ Create Block Design



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤ IP INTEGRATOR ➔ Create Block Design

➤ User IP ➔ ...

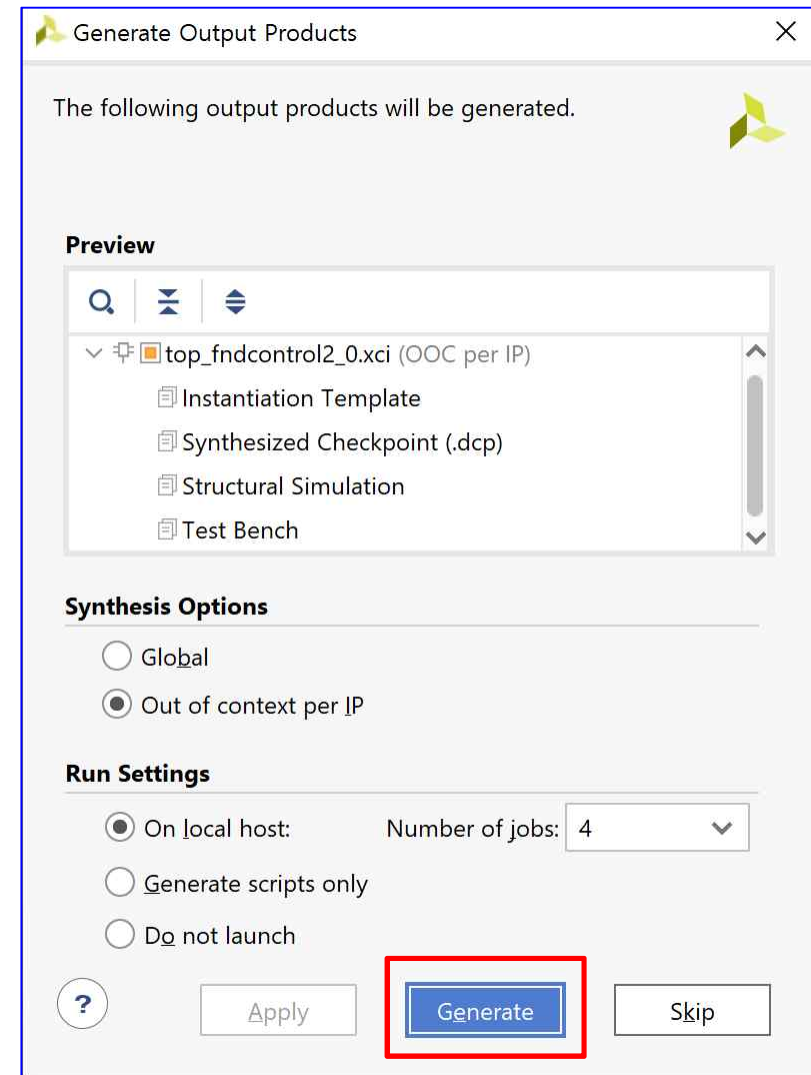
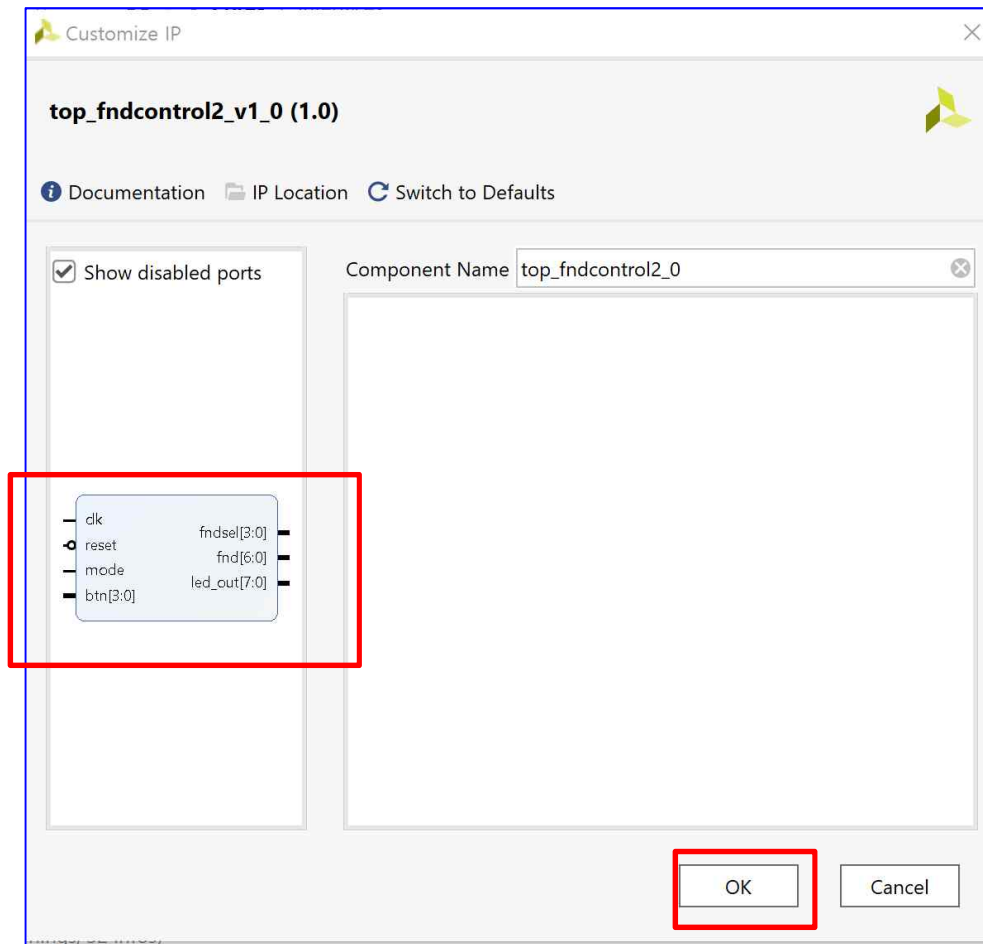
The screenshot displays the Vivado 2023.1 IP Catalog window. The left sidebar shows the 'PROJECT MANAGER' tab with 'IP Catalog' highlighted. The main window is divided into several panes:

- Sources:** Lists design sources including 'top_fndcontrol2' and its components (U0: spi_task, U1: counter, U2: clockdivider, U3: datamux4x1, U4: mux4x1).
- IP Properties:** Shows details for 'top_fndcontrol2_v1_0', including Version (1.0 Rev. 2), Description, Status (Production), License (Included), Change Log, and Vendor (Xilinx, Inc.).
- IP Catalog (2):** A table listing IP cores. The 'top_fndcontrol2_v1_0' core is highlighted in the 'User Repository' section.
- Details:** Provides a detailed view of the selected IP core, including its Name, Version, Description, Status, and License.

The 'top_fndcontrol2_v1_0' core is highlighted in the 'User Repository' section of the IP Catalog, indicating it is the selected IP for the design.

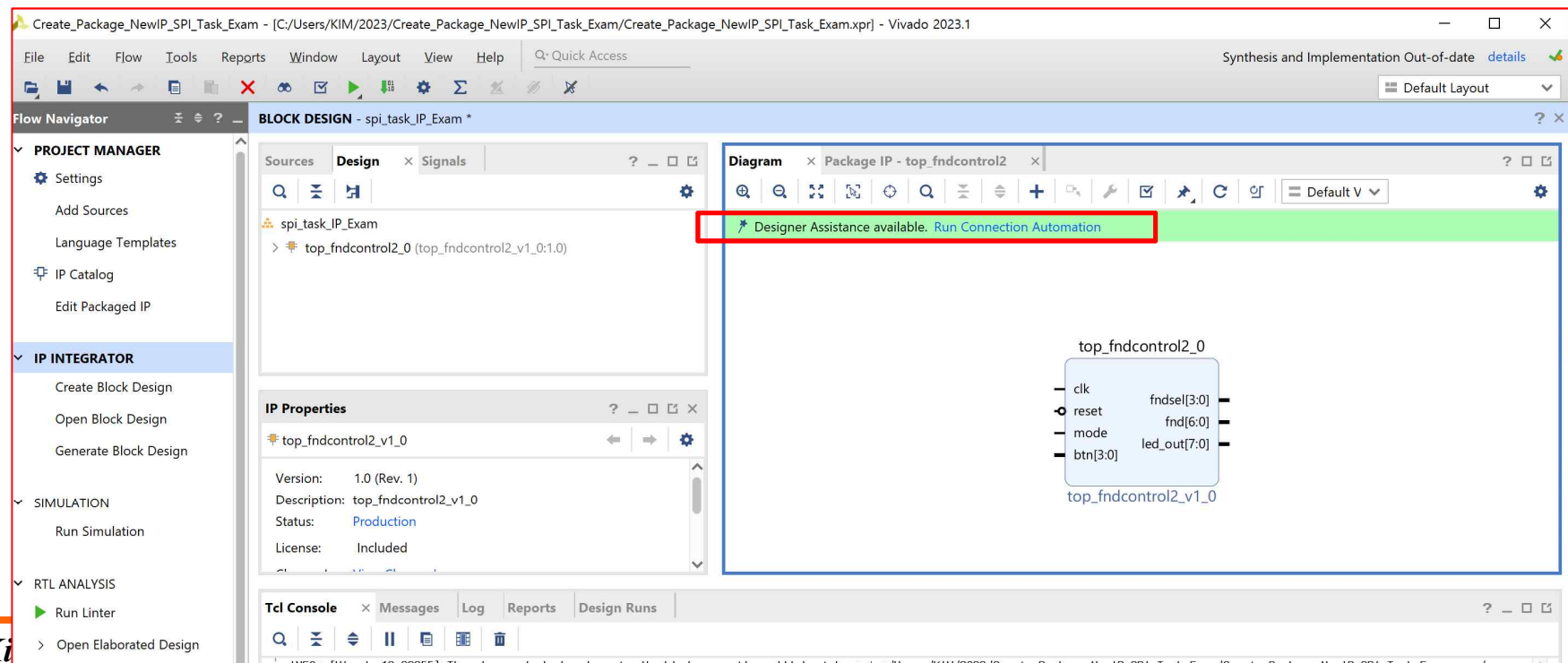
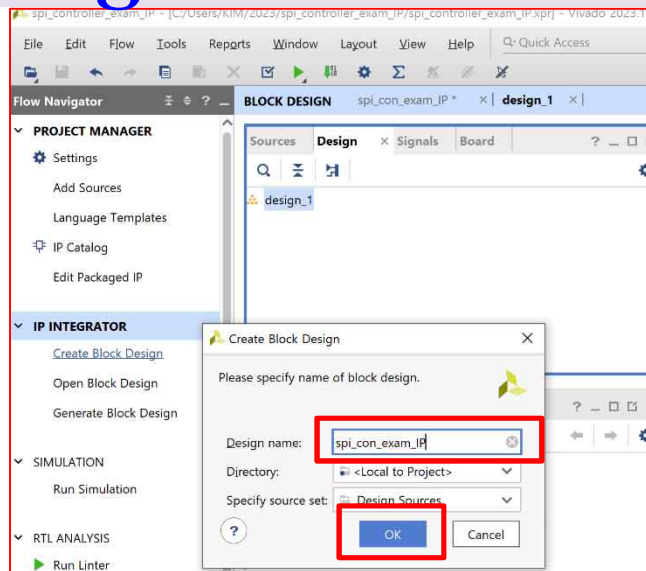
Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

The screenshot displays the Vivado 2023.1 IDE interface. The main window shows the 'BLOCK DESIGN' for 'spi_task_IP_Exam'. A red box highlights the 'Designer Assistance available. Run Connection Automation' message in the top right. The 'Run Connection Automation' dialog box is open, showing the 'All Automation' checkbox selected. The 'Description' section explains the purpose of the tool. The 'Options' section includes 'Source Clock Specification' with 'Clock Source' set to 'New Clocking Wizard' and 'Frequency MHz' set to '100'. The 'Reference Clocks' section shows 'Ref_Clk0', 'Ref_Clk1', and 'Ref_Clk2' all set to 'Auto'. The 'OK' button is highlighted with a red box.

Flow Navigator

- PROJECT MANAGER
 - Settings
 - Add Sources
 - Language Templates
 - IP Catalog
 - Edit Packaged IP
- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Run Linter
 - Open Elaborated Design

Block Design - spi_task_IP_Exam

Sources Design Signals

spi_task_IP_Exam

- top_fndcontrol2_0 (top_fndcontrol2_v1_0:1.0)

Diagram Package IP - top_fndcontrol2

top_fndcontrol2_0

clk reset mode btn[3:0] fndsel[3:0] fnd[6:0] led_out[7:0]

top_fndcontrol2_v1_0

Run Connection Automation

Automatically make connections in your design by checking the boxes of the interfaces to connect. Select an interface on the left to display its configuration options on the right.

Description

Connect clock-pin (/top_fndcontrol2_0/clk) to selected clock source. Also configure and connect clock-pins of connected bridge-IPs (AXI Interconnect, Smartconnect) as needed. Also infer Processor System Reset block and connect synchronous reset source to associated reset pin(s) as needed.

Clock: /top_fndcontrol2_0/clk

Options

Source Clock Specification

Clock Source New Clocking Wizard

Frequency MHz 100

Reference Clocks

Ref_Clk0 Auto

Ref_Clk1 Auto

Ref_Clk2 Auto

OK Cancel

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

The screenshot displays the Vivado 2023.1 IDE interface. The top menu bar includes File, Edit, Flow, Tools, Reports, Window, Layout, View, and Help. The main workspace is divided into several panes:

- Flow Navigator:** Shows the project structure with folders for Sources, Design, and Signals. The Design folder is expanded, showing the hierarchy: spi_task_IP_Exam > Nets > clk_wiz (Clocking Wizard:6.0) > rst_clk_wiz_100M (Processor System Reset:5.0) > top_fndcontrol2_0 (top_fndcontrol2_v1_0:1.0).
- Block Pin Properties:** Displays the properties for the selected block, showing the Name (clk) and Direction (Input).
- Diagram:** Shows a block diagram with the following components:
 - top_fndcontrol2_0:** A block with inputs clk, reset, mode, and btn[3:0], and outputs fndsel[3:0], fnd[6:0], and led_out[7:0].
 - clk_wiz:** A block with inputs reset and clk_in1, and outputs clk_out1 and locked.
 - rst_clk_wiz_100M:** A block with inputs slowest_sync_clk, ext_reset_in, aux_reset_in, mb_debug_sys_rst, and dcm_locked, and outputs mb_reset, bus_struct_reset[0:0], peripheral_reset[0:0], interconnect_aresetn[0:0], and peripheral_aresetn[0:0].
- Tcl Console:** Shows the Run Connection Automation command.

The **Run Connection Automation** dialog box is open, displaying the following configuration options:

- ☒ All Automation (3 out of 3 selected)
- ☒ clk_wiz
 - ☒ clk_in1
 - ☒ reset
- ☒ rst_clk_wiz_100M
 - ☒ ext_reset_in

The dialog box also includes an OK button and a Cancel button.

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

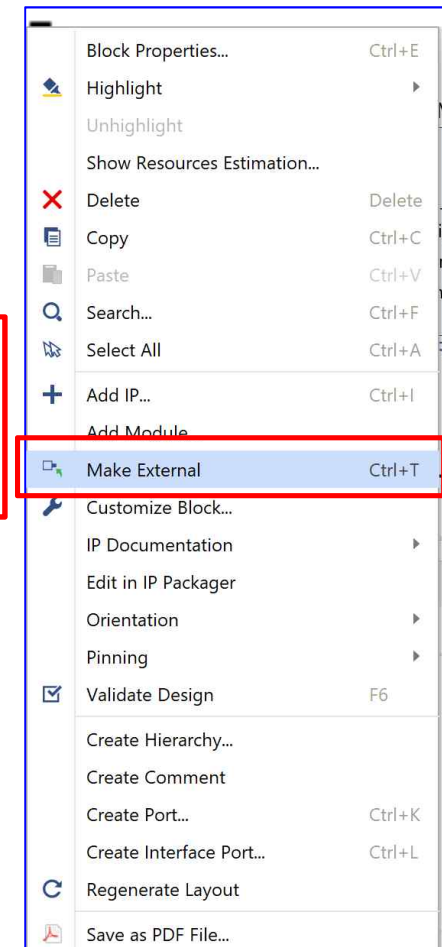
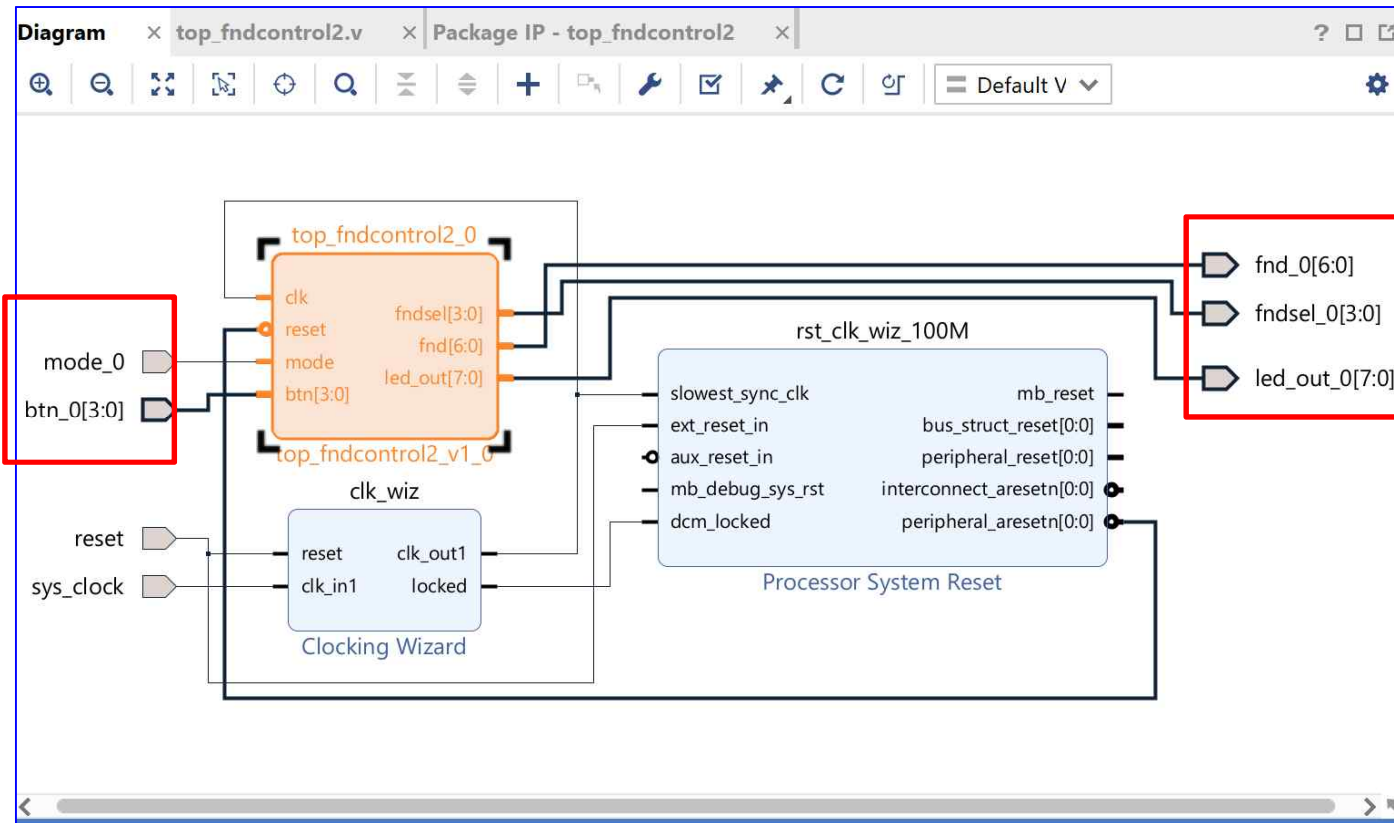
The screenshot displays the Vivado 2023.1 IDE interface. The 'IP INTEGRATOR' tab is selected in the left-hand 'Flow Navigator'. The 'BLOCK DESIGN' section shows the 'spi_con_exam_IP' project. The 'Design' tab is active, showing the 'top_fndcontrol2.v' diagram. The 'Block Properties' window for 'top_fndcontrol2_0' is open, showing the 'General' tab. The 'Name' field is set to 'top_fndcontrol2_0' and the 'Parent name' is 'spi_con_exam_IP'. The 'Tcl Console' at the bottom shows the following commands:

```
/top_fndcontrol2_0/mode  
/top_fndcontrol2_0/btn  
  
create_bd_design "design_1"  
Wrote : <C:\Users\KIM\2023\spi_controller_exam_IP\spi_controller_exam_IP.srcs\sources_1\bd\design_1\design_1.bd>  
update_compile_order -fileset sources_1  
open_bd_design {C:/Users/KIM/2023/spi_controller_exam_IP/spi_controller_exam_IP.srcs/sources_1/bd/design_1/design_1.bd}  
open_bd_design {C:/Users/KIM/2023/spi_controller_exam_IP/spi_controller_exam_IP.srcs/sources_1/bd/spi_con_exam_IP/spi_con_exam_IP.bd}
```

A context menu is open over the 'top_fndcontrol2_0' block, with the 'Make External' option highlighted. The menu includes options such as 'Block Properties...', 'Highlight', 'Unhighlight', 'Show Resources Estimation...', 'Delete', 'Copy', 'Paste', 'Search...', 'Select All', 'Add IP...', 'Add Module', 'Make External' (highlighted), 'Customize Block...', 'IP Documentation', 'Edit in IP Packager', 'Orientation', 'Pinning', 'Validate Design', 'Create Hierarchy...', 'Create Comment', 'Create Port...', 'Create Interface Port...', 'Regenerate Layout', and 'Save as PDF File...'.

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

The screenshot displays the Xilinx IDE interface with the 'Re-customize IP' dialog box open, specifically the 'Clocking Wizard (6.0)' tab. The wizard is configured for a component named 'clk_wiz'. The 'Output Clocks' tab is active, showing a table of output clocks. The 'Clocking Feedback' section is also visible, with 'Automatic Control On-Chip' selected. The 'Reset Type' is set to 'Active Low', which is highlighted with a red box. The 'Enable Optional Inputs / Outputs for MMCM/PLL' section shows 'reset' checked. The background shows a block diagram of the 'top_fndcontrol2.v' design, including a 'Clocking Wizard' block and various control signals like 'mode_0', 'btn_0[3:0]', 'reset', and 'sys_clock'.

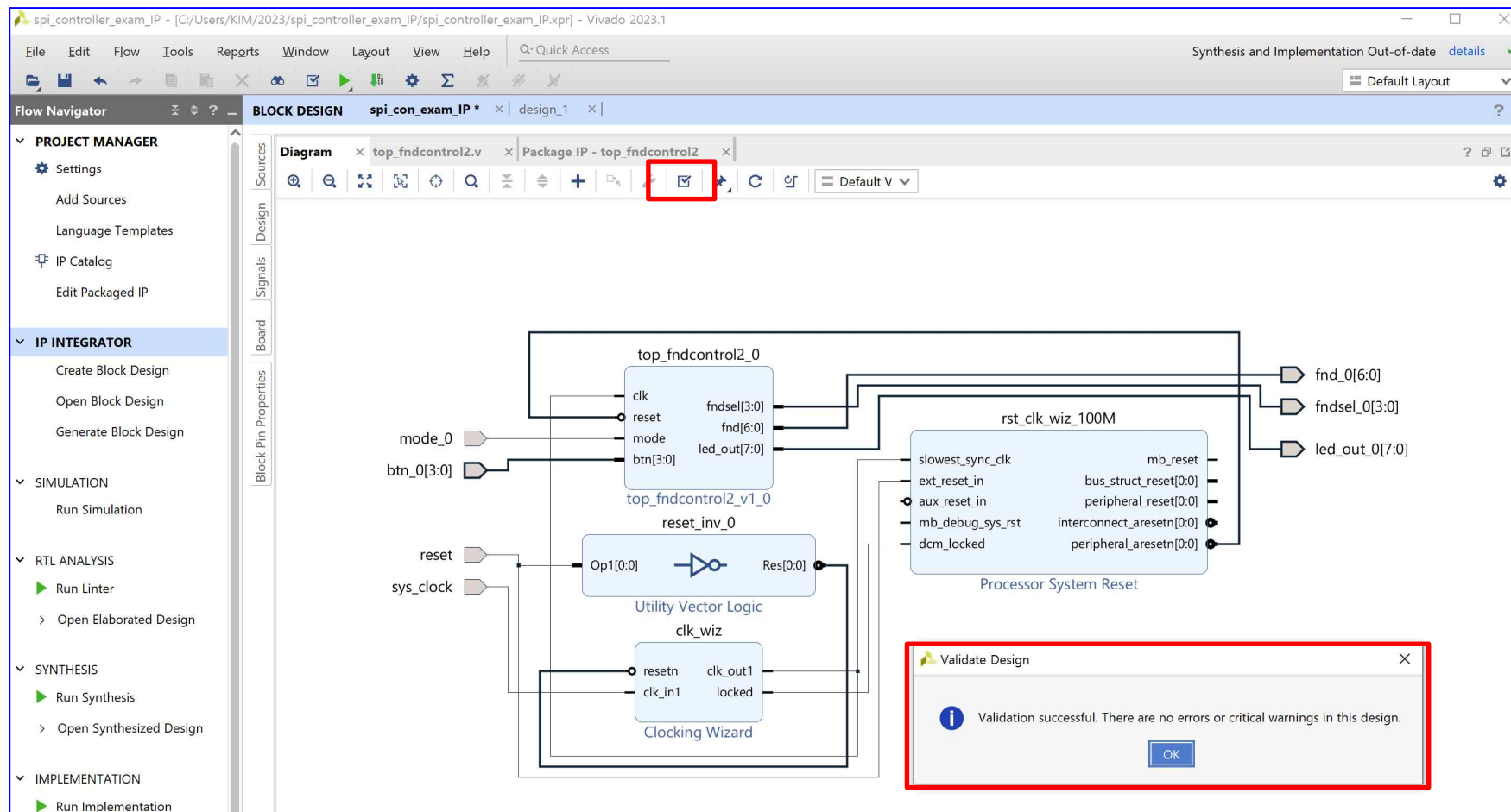
Output Clocks Table:

Output Clock	Port Name	Output Freq (MHz)	Requested	Actual	Phase (degrees)	Requested	Actual	Duty Cycle
☒ clk_out1	clk_out1	100.000	100.000000	0.000	0.000	50.000		
☐ clk_out2	clk_out2	100.000	N/A	0.000	N/A	50.000		
☐ clk_out3	clk_out3	100.000	N/A	0.000	N/A	50.000		
☐ clk_out4	clk_out4	100.000	N/A	0.000	N/A	50.000		
☐ clk_out5	clk_out5	100.000	N/A	0.000	N/A	50.000		
☐ clk_out6	clk_out6	100.000	N/A	0.000	N/A	50.000		
☐ clk_out7	clk_out7	100.000	N/A	0.000	N/A	50.000		

Reset Type: ☐ Active High ☒ Active Low

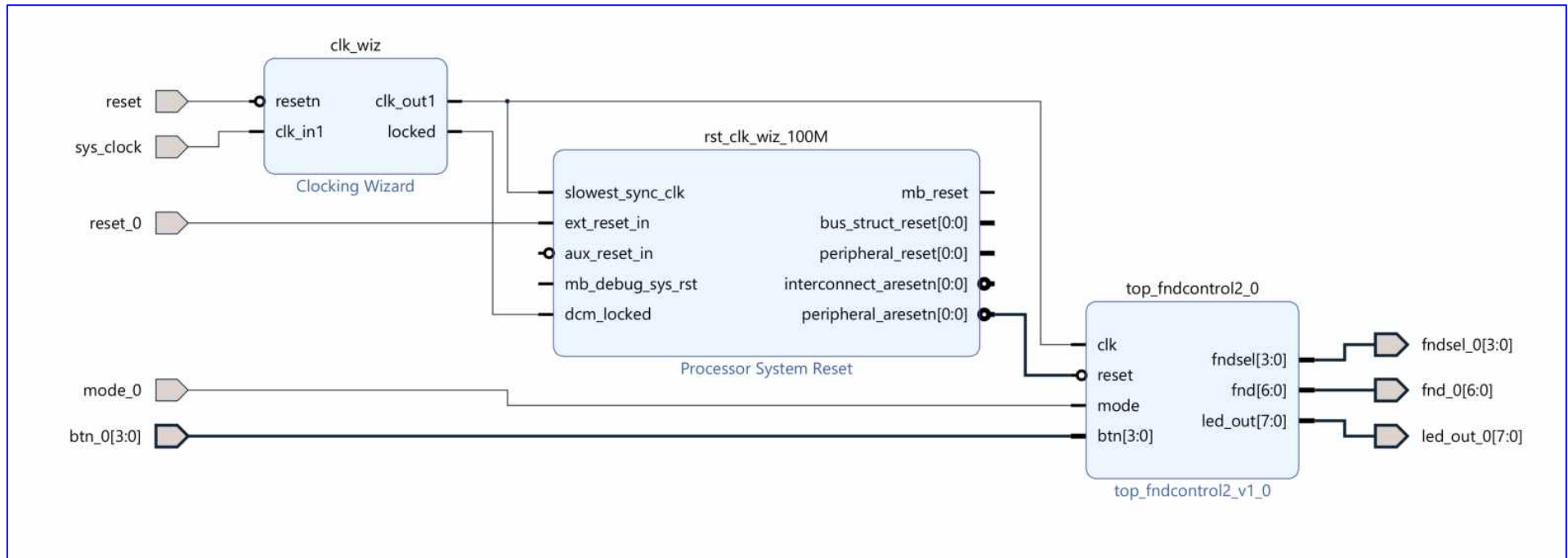
Create and Package New IP :SPI

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Creat_digital_clock_IP_exam.xpr



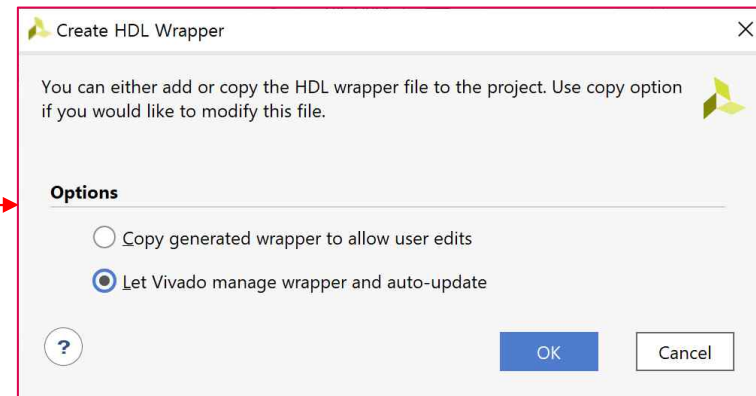
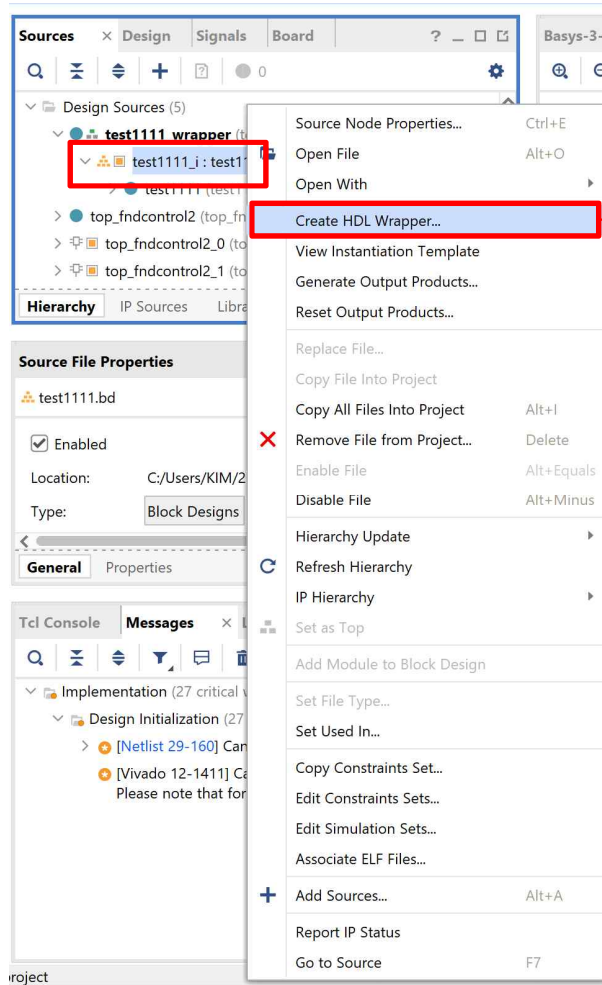
Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr



Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤ *.wrapper.v

*.wrapper.v (1)

```
1. `timescale 1 ps / 1 ps
2. module test1111_wrapper
3.     (btn_0,
4.       fnd_0,
5.       fndsel_0,
6.       led_out_0,
7.       mode_0,
8.       reset,
9.       reset_0,
10.      sys_clock);
11. input [3:0]btn_0;
12. output [6:0]fnd_0;
13. output [3:0]fndsel_0;
14. output [7:0]led_out_0;
15. input mode_0;
16. input reset;
17. input reset_0;
18. input sys_clock;
```

*.wrapper.v (2)

```
19. wire [3:0]btn_0;
20. wire [6:0]fnd_0;
21. wire [3:0]fndsel_0;
22. wire [7:0]led_out_0;
23. wire mode_0;
24. wire reset;
25. wire reset_0;
26. wire sys_clock;
27. test1111 test1111_i
28.     (.btn_0(btn_0),
29.      .fnd_0(fnd_0),
30.      .fndsel_0(fndsel_0),
31.      .led_out_0(led_out_0),
32.      .mode_0(mode_0),
33.      .reset(reset),
34.      .reset_0(reset_0),
35.      .sys_clock(sys_clock));
36. endmodule
```

*.xdc

Clock signal

```
set_property -dict { PACKAGE_PIN W5 IOSTANDARD LVCMOS33 } [get_ports sys_clock]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports sys_clock]
```

Switches

```
set_property -dict { PACKAGE_PIN U1 IOSTANDARD LVCMOS33 } [get_ports {reset_0}]
set_property -dict { PACKAGE_PIN T1 IOSTANDARD LVCMOS33 } [get_ports {mode_0}]
set_property -dict { PACKAGE_PIN R2 IOSTANDARD LVCMOS33 } [get_ports {reset}]
```

LEDs

```
set_property -dict { PACKAGE_PIN U16 IOSTANDARD LVCMOS33 } [get_ports {led_out_0[0]}]
set_property -dict { PACKAGE_PIN E19 IOSTANDARD LVCMOS33 } [get_ports {led_out_0[1]}]
set_property -dict { PACKAGE_PIN U19 IOSTANDARD LVCMOS33 } [get_ports {led_out_0[2]}]
set_property -dict { PACKAGE_PIN V19 IOSTANDARD LVCMOS33 } [get_ports {led_out_0[3]}]
set_property -dict { PACKAGE_PIN W18 IOSTANDARD LVCMOS33 } [get_ports {led_out_0[4]}]
set_property -dict { PACKAGE_PIN U15 IOSTANDARD LVCMOS33 } [get_ports {led_out_0[5]}]
set_property -dict { PACKAGE_PIN U14 IOSTANDARD LVCMOS33 } [get_ports {led_out_0[6]}]
set_property -dict { PACKAGE_PIN V14 IOSTANDARD LVCMOS33 } [get_ports {led_out_0[7]}]
```

##7 Segment Display

```
set_property -dict { PACKAGE_PIN W7 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[0]}]
set_property -dict { PACKAGE_PIN W6 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[1]}]
set_property -dict { PACKAGE_PIN U8 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[2]}]
set_property -dict { PACKAGE_PIN V8 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[3]}]
set_property -dict { PACKAGE_PIN U5 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[4]}]
set_property -dict { PACKAGE_PIN V5 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[5]}]
set_property -dict { PACKAGE_PIN U7 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[6]}]
set_property -dict { PACKAGE_PIN V7 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[7]}]
set_property -dict { PACKAGE_PIN U2 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[0]}]
set_property -dict { PACKAGE_PIN U4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[1]}]
set_property -dict { PACKAGE_PIN V4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[2]}]
set_property -dict { PACKAGE_PIN W4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[3]}]
```

##Buttons

```
#set_property -dict { PACKAGE_PIN U18 IOSTANDARD LVCMOS33 } [get_ports {resetn}]
set_property -dict { PACKAGE_PIN T18 IOSTANDARD LVCMOS33 } [get_ports {btn_0[0]}]
set_property -dict { PACKAGE_PIN W19 IOSTANDARD LVCMOS33 } [get_ports {btn_0[1]}]
set_property -dict { PACKAGE_PIN T17 IOSTANDARD LVCMOS33 } [get_ports {btn_0[2]}]
set_property -dict { PACKAGE_PIN U17 IOSTANDARD LVCMOS33 } [get_ports {btn_0[3]}]
```

Configuration options, can be used for all designs

```
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]
```

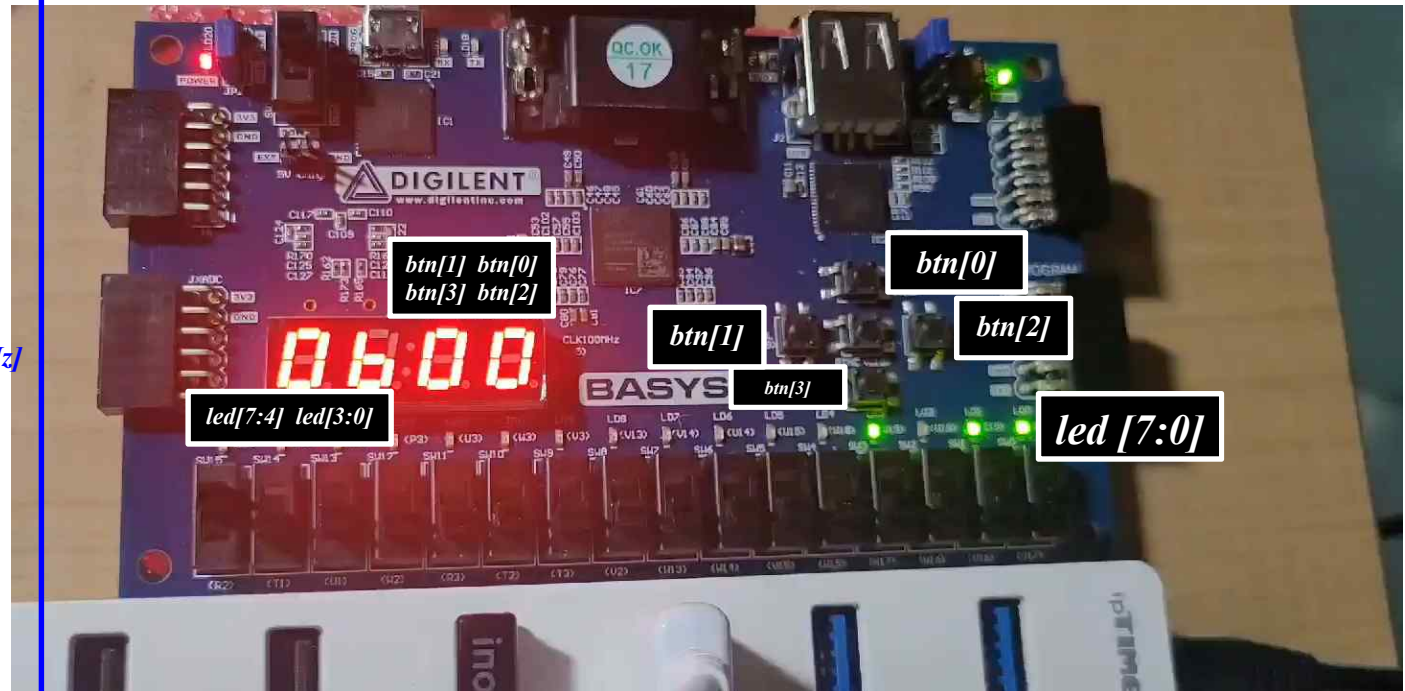
SPI configuration mode options for QSPI boot, can be used for all designs

```
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]
```

Create and Package New IP :SPI

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

- *btn_0[0]* → BTNR (T18)
- *btn_0 [1]* → BTND (W19)
- *btn_0 [2]* → BTNU (T17)
- *btn_0 [3]* → BTNU (U17)
- *reset* → SW15
- *mode_0* → SW14
- *reset_0* → SW13
- *sys_clock* → System Clock [100MHz]
- *FND4* → *rdata 10 (led[7:4])*
- *FND3* → *rdata 0(led[7:4])*
- *FND2* → Mode 0 : *btn[1]*
- *FND1* → Mode 0 : *btn[0]*
- *FND2* → Mode 1 : *btn[3]*
- *FND1* → Mode 1 : *btn[2]*
- *LD 7 ~ LD 0* → *led_out [7:0]*



[Create and Package New IP (II)]

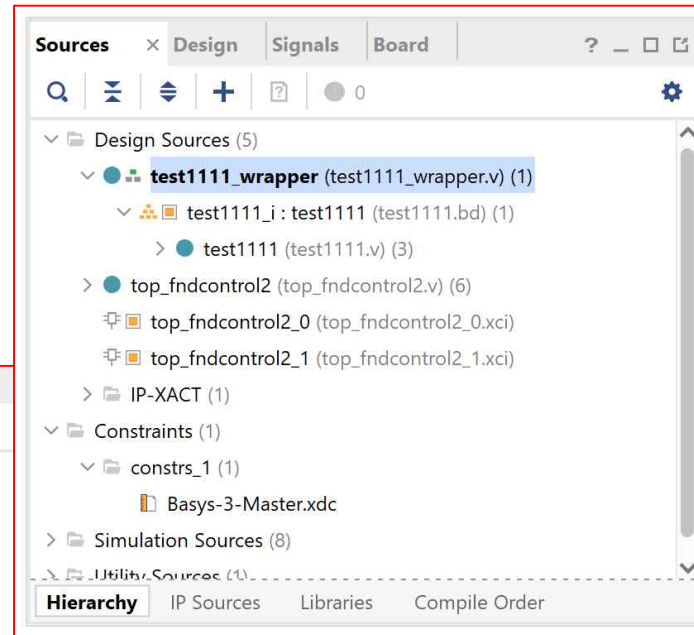
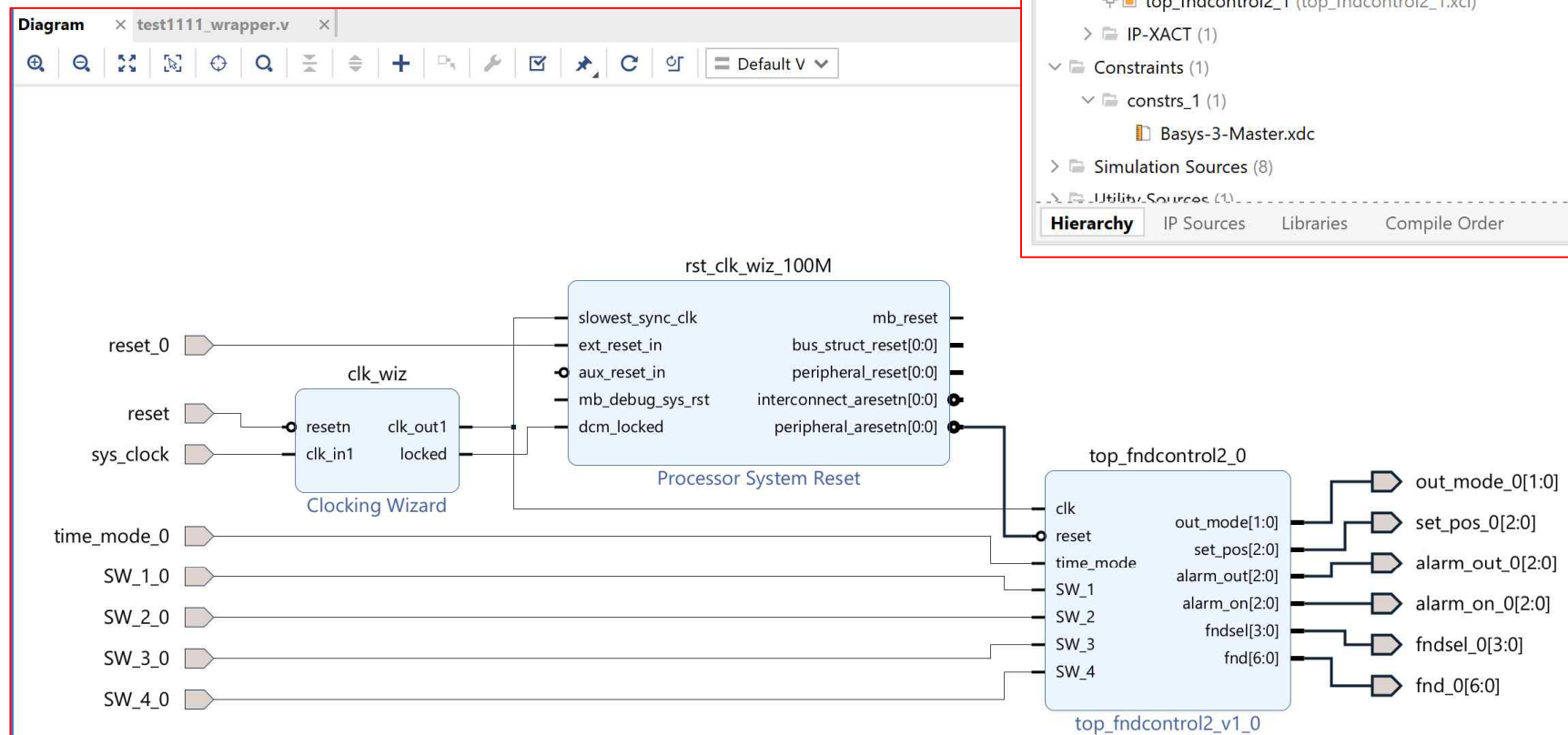
Digital_Clock and Top_FndController

➡ 실습 ⬅

Create and Package New IP : Digital Clock

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤ Source and RTL



Create and Package New IP : Digital Clock

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤ *.wrapper.v

```
Diagram x Basys-3-Master.xdc x test1111_wrapper.v x
C:/Users/KIM/2023/Creat_digital_clock_IP_exam/Creat_digital_clock_IP_exam.gen/sources_1/bd/test1111/hdl/test1111_wrapper.v

11 `timescale 1 ps / 1 ps
12
13 module test1111_wrapper
14     (SW_1_0,
15      SW_2_0,
16      SW_3_0,
17      SW_4_0,
18      alarm_on_0,
19      alarm_out_0,
20      fnd_0,
21      fndsel_0,
22      out_mode_0,
23      reset,
24      set_pos_0,
25      sys_clock,
26      time_mode_0);
27     input SW_1_0;
28     input SW_2_0;
29     input SW_3_0;
30     input SW_4_0;
31     output [2:0]alarm_on_0;
32     output [2:0]alarm_out_0;
33     output [6:0]fnd_0;
34     output [3:0]fndsel_0;
35     output [1:0]out_mode_0;
36     input reset;
37     output [2:0]set_pos_0;
38     input sys_clock;
39     input time_mode_0;
40
41     wire SW_1_0;
```

```
Diagram x Basys-3-Master.xdc x test1111_wrapper.v x
C:/Users/KIM/2023/Creat_digital_clock_IP_exam/Creat_digital_clock_IP_exam.gen/sources_1/bd/test1111/hdl/test1111_wrapper.v

40
41     wire SW_1_0;
42     wire SW_2_0;
43     wire SW_3_0;
44     wire SW_4_0;
45     wire [2:0]alarm_on_0;
46     wire [2:0]alarm_out_0;
47     wire [6:0]fnd_0;
48     wire [3:0]fndsel_0;
49     wire [1:0]out_mode_0;
50     wire reset;
51     wire [2:0]set_pos_0;
52     wire sys_clock;
53     wire time_mode_0;
54
55     test1111 test1111_i
56         (.SW_1_0(SW_1_0),
57          .SW_2_0(SW_2_0),
58          .SW_3_0(SW_3_0),
59          .SW_4_0(SW_4_0),
60          .alarm_on_0(alarm_on_0),
61          .alarm_out_0(alarm_out_0),
62          .fnd_0(fnd_0),
63          .fndsel_0(fndsel_0),
64          .out_mode_0(out_mode_0),
65          .reset(reset),
66          .set_pos_0(set_pos_0),
67          .sys_clock(sys_clock),
68          .time_mode_0(time_mode_0));
69 endmodule
70
```


Create and Package New IP : Digital Clock

Creat_Package_NewIP_SPI_Task_Exam.xpr
Creat_digital_clock_IP_exam.xpr

➤ .XDC

*.xdc

Clock signal

```
set_property -dict { PACKAGE_PIN W5 IOSTANDARD LVCMOS33 } [get_ports sys_clock]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports sys_clockk]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets SW_1_IBUF]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets SW_2_IBUF]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets SW_3_IBUF]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets SW_4_IBUF]
```

Switches

```
#set_property -dict { PACKAGE_PIN V17 IOSTANDARD LVCMOS33 } [get_ports {SW_1}]
#set_property -dict { PACKAGE_PIN V16 IOSTANDARD LVCMOS33 } [get_ports {SW_2}]
#set_property -dict { PACKAGE_PIN W16 IOSTANDARD LVCMOS33 } [get_ports {SW_3}]
#set_property -dict { PACKAGE_PIN W17 IOSTANDARD LVCMOS33 } [get_ports {inx[3]}]
#set_property -dict { PACKAGE_PIN W15 IOSTANDARD LVCMOS33 } [get_ports {iny[0]}]
#set_property -dict { PACKAGE_PIN V15 IOSTANDARD LVCMOS33 } [get_ports {iny[1]}]
#set_property -dict { PACKAGE_PIN W14 IOSTANDARD LVCMOS33 } [get_ports {iny[2]}]
#set_property -dict { PACKAGE_PIN W13 IOSTANDARD LVCMOS33 } [get_ports {iny[3]}]
#set_property -dict { PACKAGE_PIN V2 IOSTANDARD LVCMOS33 } [get_ports {inc[0]}]
#set_property -dict { PACKAGE_PIN T3 IOSTANDARD LVCMOS33 } [get_ports {inc[1]}]
#set_property -dict { PACKAGE_PIN T2 IOSTANDARD LVCMOS33 } [get_ports {inc[2]}]
#set_property -dict { PACKAGE_PIN R3 IOSTANDARD LVCMOS33 } [get_ports {inc[3]}]
#set_property -dict { PACKAGE_PIN W2 IOSTANDARD LVCMOS33 } [get_ports {fndin[0]}]
#set_property -dict { PACKAGE_PIN U1 IOSTANDARD LVCMOS33 } [get_ports {fndin[1]}]
set_property -dict { PACKAGE_PIN T1 IOSTANDARD LVCMOS33 } [get_ports {time_mode_0}]
set_property -dict { PACKAGE_PIN R2 IOSTANDARD LVCMOS33 } [get_ports {reset}]
```

LEDs

```
set_property -dict { PACKAGE_PIN U16 IOSTANDARD LVCMOS33 } [get_ports {alarm_on_0[0]}]
set_property -dict { PACKAGE_PIN E19 IOSTANDARD LVCMOS33 } [get_ports {alarm_on_0[1]}]
set_property -dict { PACKAGE_PIN U19 IOSTANDARD LVCMOS33 } [get_ports {alarm_on_0[2]}]
set_property -dict { PACKAGE_PIN V19 IOSTANDARD LVCMOS33 } [get_ports {alarm_out_0[0]}]
set_property -dict { PACKAGE_PIN W18 IOSTANDARD LVCMOS33 } [get_ports {alarm_out_0[1]}]
set_property -dict { PACKAGE_PIN U15 IOSTANDARD LVCMOS33 } [get_ports {alarm_out_0[2]}]
#set_property -dict { PACKAGE_PIN U14 IOSTANDARD LVCMOS33 } [get_ports {alarm_out[2]}]
#set_property -dict { PACKAGE_PIN V14 IOSTANDARD LVCMOS33 } [get_ports {fndnumber[2]}]
set_property -dict { PACKAGE_PIN V13 IOSTANDARD LVCMOS33 } [get_ports {out_mode_0[0]}]
set_property -dict { PACKAGE_PIN V3 IOSTANDARD LVCMOS33 } [get_ports {out_mode_0[1]}]
#set_property -dict { PACKAGE_PIN W3 IOSTANDARD LVCMOS33 } [get_ports {fndnumber[5]}]
#set_property -dict { PACKAGE_PIN U3 IOSTANDARD LVCMOS33 } [get_ports {fndnumber[6]}]
#set_property -dict { PACKAGE_PIN P3 IOSTANDARD LVCMOS33 } [get_ports {fndnumber[7]}]
set_property -dict { PACKAGE_PIN N3 IOSTANDARD LVCMOS33 } [get_ports {set_pos_0[0]}]
set_property -dict { PACKAGE_PIN P1 IOSTANDARD LVCMOS33 } [get_ports {set_pos_0[1]}]
set_property -dict { PACKAGE_PIN L1 IOSTANDARD LVCMOS33 } [get_ports {set_pos_0[2]}]
```

*.xdc

##7 Segment Display

```
set_property -dict { PACKAGE_PIN W7 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[0]}]
set_property -dict { PACKAGE_PIN W6 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[1]}]
set_property -dict { PACKAGE_PIN U8 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[2]}]
set_property -dict { PACKAGE_PIN V8 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[3]}]
set_property -dict { PACKAGE_PIN U5 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[4]}]
set_property -dict { PACKAGE_PIN V5 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[5]}]
set_property -dict { PACKAGE_PIN U7 IOSTANDARD LVCMOS33 } [get_ports {fnd_0[6]}]
```

```
#set_property -dict { PACKAGE_PIN V7 IOSTANDARD LVCMOS33 } [get_ports fnd[7]]
```

```
set_property -dict { PACKAGE_PIN U2 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[0]}]
set_property -dict { PACKAGE_PIN U4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[1]}]
set_property -dict { PACKAGE_PIN V4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[2]}]
set_property -dict { PACKAGE_PIN W4 IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[3]}]
```

##Buttons

```
#set_property -dict { PACKAGE_PIN U18 IOSTANDARD LVCMOS33 } [get_ports btnC]
set_property -dict { PACKAGE_PIN T18 IOSTANDARD LVCMOS33 } [get_ports SW_3_0]
set_property -dict { PACKAGE_PIN W19 IOSTANDARD LVCMOS33 } [get_ports SW_4_0]
set_property -dict { PACKAGE_PIN T17 IOSTANDARD LVCMOS33 } [get_ports SW_1_0]
set_property -dict { PACKAGE_PIN U17 IOSTANDARD LVCMOS33 } [get_ports SW_2_0]
```

Configuration options, can be used for all designs

```
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]
```

SPI configuration mode options for QSPI boot, can be used for all designs

```
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]
```

Create and Package New IP : Digital Clock

Creat_Package_NewIP_SPI_Task_Exam.xpr

Creat digital clock IP exam.xpr

[표 6-3] 디지털 시계의 입출력 장치

입출력	입출력 장치
clk	On board 1MHz OSC
reset	리셋 스위치
SW0~SW3	입력 키 스위치 Key0~Key3
시, 분, 초	FND3~FND1
알람 ON/OFF 상태	LED D1~D3
알람 신호	LED D5~D7
모드	LED D8~D9
시, 분, 초 설정 위치 표시	LED D13~D15

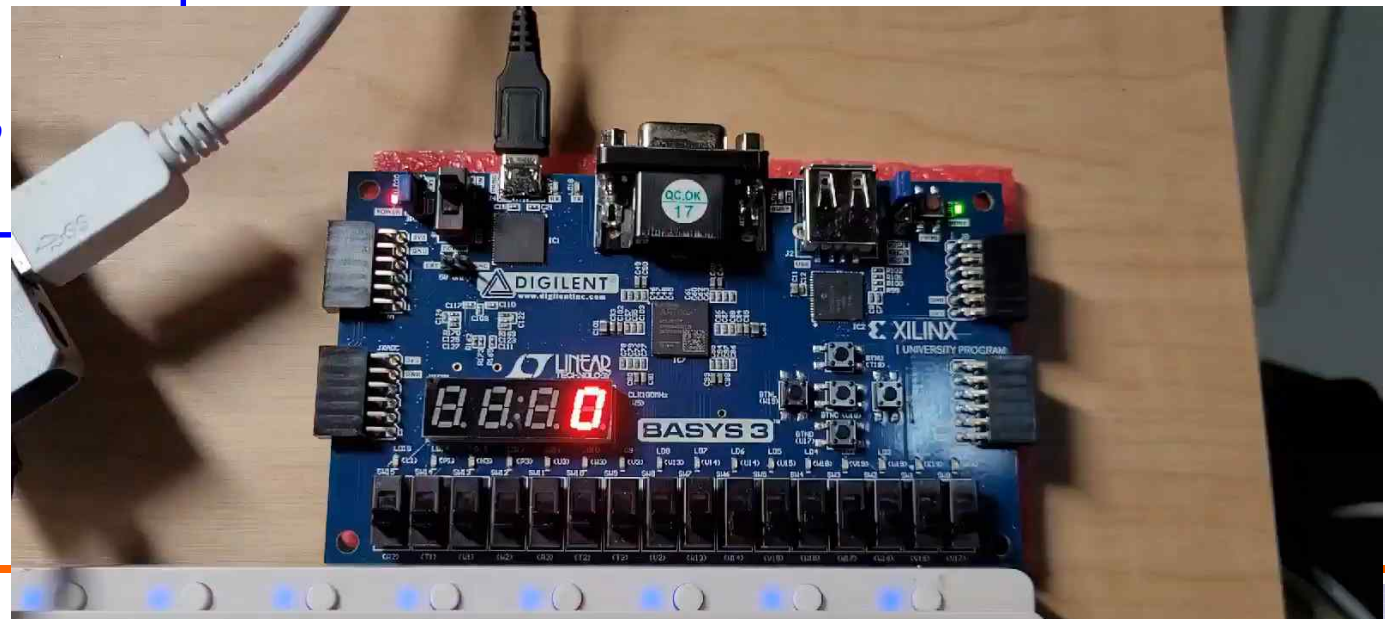
➤ Program Device

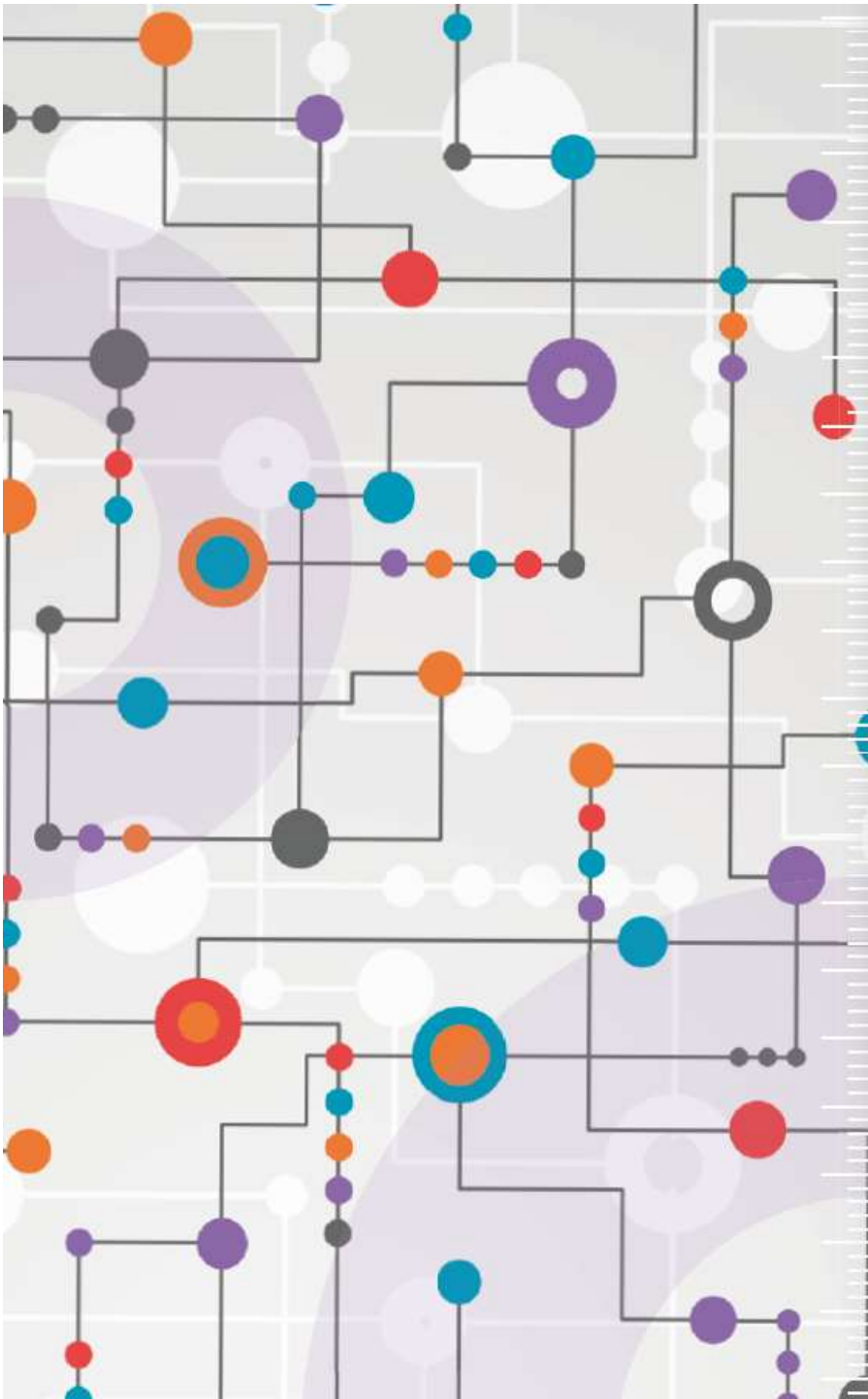
- SW0 → BTNR (T17) : Mode Selection
- SW1 → BTND (U17) : Set_positon
- SW2 → BTNU (T18) : Time Up
- SW3 → BTNL (W19) : Alam On/Off
- reset → SW15
- FND4 → Hour 10 (or Min10)
- FND3 → Hour 0 (or Min0)
- FND2 → Min 10 (or Sec10)
- FND1 → Min 0 (or Sec0)
- **time_mode → SW14 : 시분 or 분초 모드**
- Alarm On/Off → LD 0 ~ LD 2
- Alarm Signal → LD 4 ~ LD 6
- Mode(Selection) → LD 8 ~ LD 9
- Set_potion → LD 13 ~ LD 15

[표 6-1] 디지털 시계의 스위치와 동작 모드

모드 선택(SW0)	선택 1(SW1)	선택 2(SW2)	선택 3(SW3)
0	시계 모드	현재 시간 출력	
1	시간 설정	위치 선택 (Hour, Min)	시간 증가
2	알람 설정	위치 선택 (Hour, Min)	시간 증가
3	타이머(스톱워치)	스타트/스톱 (Start/Stop)	리셋

알람 ON/OFF





수고하셨습니다.